Recurrence-Free Dyadic Values

of the Fabius and Rvachev Functions

Finite moment elimination, binary compression, integer determinants, and exact quarter-base spline extraction

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ProveIt formal-development synthesis

4 September 2026

Abstract

Exact rationality of the Fabius distribution function F_{bd} and of Rvachev's up-function at dyadic arguments is classical, but the formulas that display it usually leave an auxiliary sequence — the even moments, or the values at reciprocal powers of two — defined only by a triangular recurrence, or reach the value as the limit of finite convolutions. This document removes both dependencies and organizes every finite route to the value in one place.

The principal formula expresses $F_{\mathrm{bd}}(m/2^n)$ through $\lfloor n/2 \rfloor + 1$ explicit Thue–Morse power sums with rational interpolation weights at the quarter base $q = \frac{1}{4}$. Behind it is an exact, not asymptotic, statement: at a fixed dyadic point of depth n the centred finite-uniform distribution functions are polynomials of degree at most $\lfloor n/2 \rfloor$ in 4^{-N}, and a shrinking-cell version identifies the whole polynomial piece of the finite spline with a finite deconvolution of the dyadic Taylor polynomial. The degree is exactly $\lfloor n/2 \rfloor$ at every reduced interior dyadic, so the sample count is optimal for any linear rule whose weights work on the whole grid.

A second family compresses the classical Thue–Morse moment formula to one outer summand per nonzero binary digit of the numerator — in two different ways, a block identity and a telescope — and supplies the coefficients by positive ordered-composition sums, by Bernoulli multiplicity sums with the Bernoulli numbers themselves given as finite sums, and by interpolation of finitely many uniform convolutions. An integer bordered determinant packages the value as a single ratio of integers and yields an explicit common denominator, which in turn gives a certified rounding formula. The same finite algebra evaluates the bump, the signed global extension, every dyadic derivative, all iterated primitives, and the analogous normalized reciprocal-integer-base laws. Every result carries a complete proof; every printed rational is checked by the retained exact-arithmetic verifier; the relation to the canonical extraction volume of this subgroup is stated formula by formula. No Lean formalization of the new material is claimed.

Scope. The bounded distribution function, the compactly supported bump, and the signed global extension are kept distinct throughout. The results are mathematical derivations with complete proofs; the formalization register at the end records which pieces have Lean counterparts in the repository and which do not.

Contents

1 The question and the map of formulas

1.1 What counts as a closed form here

We seek the exact values of the Fabius distribution function and of Rvachev's bump at $m/2^n$, with integer m and nonnegative integer n. A formula is called *recurrence-free* when its right-hand side contains only explicitly bounded finite sums and products of specified numbers, together with rational arithmetic, integer powers, factorials, binary digits, and finite q-Pochhammer symbols. A coefficient-extraction symbol may abbreviate a finite sum that is displayed separately, and a determinant is admissible because its entries are given explicitly and its Leibniz expansion is a finite sum. What is excluded is a definition that leaves a desired auxiliary quantity specified only in terms of earlier unknown quantities: a formula that refers to "the moments c_k, computed recursively" is not a closed form in this sense, and neither is a limit of explicit rational approximations. Recursion may of course be an efficient *implementation*; the point is that no recurrence is needed to *define* the numbers on the right-hand side of an evaluation identity.

Three functions must be distinguished. Throughout, F_{bd} is the bounded Fabius distribution function, up is the compactly supported bump, and F_{gl} is the signed global extension. On their principal domains,

$$\mathrm{up}(x) = F_{\mathrm{bd}}(1-|x|), \qquad F_{\mathrm{bd}}(x) = \mathrm{up}(x-1) \quad (0 \le x \le 1), \qquad F_{\mathrm{gl}} = F_{\mathrm{bd}} \text{ on } [0,1]. \tag{1}$$

The precise conventions and the probability model are fixed in Section 2.

1.2 The principal formula

Write

$$T_r := \frac{r(r+1)}{2}, \qquad d := \lfloor n/2 \rfloor, \qquad \varepsilon_h := (-1)^{\mathrm{wt}_2(h)}, \tag{2}$$

where $\mathrm{wt}_2(h)$ is the number of ones in the binary expansion of h. The Thue–Morse sign is a digit formula, not a recursively defined sequence: for $0 \le h < 2^L$ its digits are the explicit integers $\lfloor h/2^\ell \rfloor - 2\lfloor h/2^{\ell+1} \rfloor$, $0 \le \ell < L$, and

$$\varepsilon_h = \prod_{\ell=0}^{L-1} \Big(1 - 2\big(\big\lfloor h/2^\ell \big\rfloor - 2\big\lfloor h/2^{\ell+1} \big\rfloor\big)\Big). \tag{3}$$

Define the *integer Thue–Morse power sum*

$$\mathcal{S}_N(A) := \sum_{h=0}^{A-1} \varepsilon_h \,(2A - 2h - 1)^N \qquad (N \ge 1,\ A \ge 0). \tag{4}$$

Empty sums are zero, empty products are one, and $0^0 = 1$ in the finite polynomial formulas.

Theorem 1.1 (Quarter-base closed formula)**.** *Let $n \ge 1$, $0 \le m \le 2^n$, and $d = \lfloor n/2 \rfloor$. Put*

$$w_{d,j} := \frac{(-1)^{d-j}\, 4^{T_j}}{\displaystyle\prod_{r=1}^{j}(4^r - 1) \prod_{r=1}^{d-j}(4^r - 1)} \qquad (0 \le j \le d). \tag{5}$$

Then

$$\boxed{F_{\mathrm{bd}}\Big(\frac{m}{2^n}\Big) = \sum_{j=0}^{d} w_{d,j}\, \frac{\mathcal{S}_{n+j}(2^j m)}{2^{T_{n+j}}\,(n+j)!}\,.} \tag{6}$$

More generally, for every integer $R \geq 0$,

$$F_{\mathrm{bd}}\left(\frac{m}{2^n}\right) = \sum_{j=0}^{d} w_{d,j} \, \frac{\mathcal{S}_{n+R+j}(2^{R+j}m)}{2^{T_{n+R+j}}\,(n+R+j)!}\,. \tag{7}$$

The proof is completed in Section 8 (Theorem 8.3). It does *not* assume rationality of the target value, so Equation (6) is itself a proof of dyadic rationality using only rational finite expressions. The formula contains no moments, Bernoulli numbers, derivative values, recursively defined coefficients, or limits. The first rows of weights are

$$\begin{array}{c|l} d & (w_{d,0},\ldots,w_{d,d}) \\ \hline 0 & (1) \\ 1 & (-1,\ 4)/3 \\ 2 & (1,\ -20,\ 64)/45 \\ 3 & (-1,\ 84,\ -1344,\ 4096)/2835 \end{array} \tag{8}$$

Each row sums to one. These are exact interpolation rows, not fitted numerical coefficients.

1.3 The families of formulas

There are several genuinely different finite routes to the same rational number, and they are useful in different situations. Table 1 lists them with their characteristic ingredient.

Family	Finite ingredients and characteristic feature	Main result
Bernoulli partitions	multiplicity vectors and finite Bernoulli power sums; visible cumulant structure	Theorem 4.5
Positive compositions	ordered positive parts and Mersenne-type factors $4^S - 1$; every coefficient term is positive	Theorem 4.8
Binary blocks	one outer summand per nonzero binary digit; centred coefficients; valid for the signed extension at every numerator	Theorem 5.1
Binary telescope	the same digits, an alternating telescope of raw-moment polynomials	Theorem 5.4
Finite uniform prefixes	moments of finitely many uniforms, interpolated at the quarter base	Theorem 6.1, Theorem 6.2
Integer determinant	a bordered matrix of order $\lfloor n/2 \rfloor + 1$; integer certificate and a common denominator	Theorem 10.1
Certified rounding	one enormous prefix, rounded to the common denominator	Theorem 10.5
Quarter-base extraction	$\lfloor n/2 \rfloor + 1$ centred spline levels; no auxiliary coefficients at all	Theorem 1.1
Direct bump extraction	the same weights applied to finite densities	Theorem 11.1

Table 1: The families of recurrence-free formulas. All rows are exact.

For a single value with a sparse binary expansion, one of the two binary formulas usually avoids the largest sums. For many values on one grid, the master formula with centred coefficients permits the same explicitly computed coefficients to be reused. The composition expression is transparent when positive terms and elementary denominators are wanted; the partition expression connects to cumulants and suits symbolic manipulation; the determinant is a compact integer certificate with a visible denominator. The quarter-base formula is the most self-contained answer when *no auxiliary coefficient sequence at all* is acceptable: it contains only a short geometric interpolation row and finite power sums. Section 14 compares them further.

1.4 Relation to the canonical volume and to the literature

This document belongs to the subgroup whose canonical volume [2] proves that, at a dyadic point x of depth s, the finite sinc-product spline up_n obeys $\mathrm{up}_n(x) = \mathrm{up}(x) + \sum_{r=1}^{\lfloor s/2 \rfloor} C_r(x) 4^{-rn}$ exactly for all $n \geq s$, and derives from it the quarter-base Gaussian-binomial extraction row. The present document proves the same polynomiality in the distribution-function normalization (Theorems 7.3 and 7.4) and reads the row as an *evaluation formula*, Equation (6), in which the spline values themselves are replaced by explicit integer sums. Everything else here — the moment-free coefficients, the binary compressions, the determinant, the denominators, the shifted half-base identity, the primitives, and the integer-base extension — is complementary to the volume. Section 8.5 gives the dictionary between the two normalizations, and Section D the formalization register.

The classical background is Arias de Reyna's arithmetic and analytic papers [4, 5], which contain the dyadic moment formula and the sharp denominator theory; Haugland [6] gives an exact dyadic evaluator; Reshetnikov's 2019 question [7] posed the half-base q-binomial expression recovered in Section 9; and Pinelis [8] gave a finite-convolution representation followed by a limit, which is exactly the limiting step that the quarter-base row removes. The repository's primary exposition [1] already contains the Thue–Morse moment formula, the reciprocal-power path and composition descriptions, the Bell and cumulant machinery, and the half-base layer; the Exponents volume [3] contains the asymptotic quarter-base Richardson combination. None of those is claimed as new. What this document adds is the completion of every coefficient into a finite expression, the organization of the distinct evaluation routes with full proofs, and the exact centred cutoff identity with its precise degree.

1.5 Conventions

Throughout, n, m, N are nonnegative integers unless another range is stated, and $\mathbb{N}_0 = \{0, 1, 2, \ldots\}$. For $k \geq 0$, $\mathcal{C}_k$ is the finite set of ordered compositions of k into positive parts, the empty composition being the only element of $\mathcal{C}_0$; there are 2^{k-1} compositions of $k \geq 1$, obtained by choosing cuts among $k-1$ gaps. For $\mathbf{r} = (r_1, \ldots, r_\ell) \in \mathcal{C}_k$ the *suffix sums* are

$$R_i := r_i + r_{i+1} + \cdots + r_\ell \qquad (1 \leq i \leq \ell), \tag{9}$$

so that $R_1 = k$ and the R_i are distinct integers in $[1, k]$. The *multiplicity-profile set* is

$$\mathcal{P}_k := \left\{ (v_1, \ldots, v_k) \in \mathbb{N}_0^k : v_1 + 2v_2 + \cdots + kv_k = k \right\}, \tag{10}$$

which for $k = 0$ consists of the empty tuple. For $0 < q < 1$,

$$(q; q)_r = \prod_{j=1}^{r} (1 - q^j), \qquad \begin{bmatrix} d \\ j \end{bmatrix}_q = \frac{(q; q)_d}{(q; q)_j \, (q; q)_{d-j}}, \tag{11}$$

finite products; at $q = \frac{1}{4}$ every such value is rational. No infinite q-product is needed to evaluate any dyadic value in this document. Bernoulli numbers use the convention $B_1 = -\frac{1}{2}$. A table of the symbols is in Section C.

2 Normalization and the probability model

2.1 The bounded distribution function and its centred density

Let $U_1, U_2, \ldots$ be independent random variables, each uniform on $[0, 1]$, and define

$$X := \sum_{j=1}^{\infty} 2^{-j} U_j, \qquad F_{\mathrm{bd}}(x) := \mathbb{P}(X \leq x). \tag{12}$$

The series converges absolutely for every outcome, since $0 \le U_j \le 1$ and $\sum_j 2^{-j} = 1$, and takes values in $[0, 1]$. Let $V_j = 2U_j - 1$, uniform on $[-1, 1]$, and

$$Z := 2X - 1 = \sum_{j=1}^{\infty} 2^{-j} V_j. \tag{13}$$

The density of Z is denoted up; this is the standard probability construction of the two functions [4, 5], and the catalogue entry `rvachev-up` normalizes up in exactly this way.

Lemma 2.1 (Smoothness and normalization)**.** *The density* up *is even, nonnegative, C^∞, and supported on $[-1, 1]$. Moreover*

$$F'_{\mathrm{bd}}(x) = 2\,\mathrm{up}(2x - 1), \tag{14}$$
$$\mathrm{up}(x) = F_{\mathrm{bd}}(1 - |x|), \tag{15}$$
$$F_{\mathrm{bd}}(1 - x) = 1 - F_{\mathrm{bd}}(x), \tag{16}$$
$$\mathrm{up}'(x) = 2\,\mathrm{up}(2x + 1) - 2\,\mathrm{up}(2x - 1), \tag{17}$$
$$\mathrm{up}(0) = 1, \qquad 0 \le \mathrm{up} \le 1. \tag{18}$$

The function F_{bd} is C^∞ on $\mathbb{R}$, equals 0 for $x \le 0$ and 1 for $x \ge 1$, is 2-Lipschitz, and all its positive-order derivatives vanish at 0 and at 1. All derivatives of up *vanish at ± 1.*

Proof. The law of Z is symmetric because every V_j has a symmetric law, and it has no atoms: conditioning on all coordinates but U_1 leaves a translated uniform distribution. Its characteristic function is

$$\mathbb{E}\, \mathrm{e}^{\mathrm{i}tZ} = \prod_{j \ge 1} \frac{\sin(t/2^j)}{t/2^j}.$$

For any fixed positive integer L, bound the first L factors by $\min(1, 2^j / |t|)$ and all remaining factors by one: this gives decay $\mathcal{O}_L\big((1 + |t|)^{-L}\big)$. Hence the characteristic function multiplied by any fixed power of t is integrable, and Fourier inversion with differentiation under the integral gives a C^∞ density. Nonnegativity and the support $[-1, 1]$ follow from the law of the bounded sum; the affine change $Z = 2X - 1$ gives Equation (14).

Replacing every U_j by $1 - U_j$ shows $X \overset{\mathrm{law}}{=} 1 - X$, which is Equation (16). For Equation (15) use the independent-copy decomposition $Z \overset{\mathrm{law}}{=} (V + Z')/2$, with V uniform on $[-1, 1]$ and Z' an independent copy of Z. The density of $V/2$ is one on $[-\frac{1}{2}, \frac{1}{2}]$, so

$$\mathrm{up}(x) = \mathbb{P}(2x - 1 \le Z' \le 2x + 1) = F_{\mathrm{bd}}(x + 1) - F_{\mathrm{bd}}(x). \tag{19}$$

For $x \ge 0$ the upper endpoint is at least one, so the probability equals $\mathbb{P}(X' \ge x) = 1 - F_{\mathrm{bd}}(x) = F_{\mathrm{bd}}(1 - x)$; for $x \le 0$ use evenness. The same display applies outside the support through the constant tails of F_{bd}. Differentiating Equation (19) and using Equation (14) gives Equation (17). Formula Equation (19) gives $\mathrm{up}(0) = F_{\mathrm{bd}}(1) - F_{\mathrm{bd}}(0) = 1$ and $\mathrm{up} \le 1$, and Equation (14) then bounds F'_{bd} by two. Smoothness together with the constant exterior values forces every positive-order derivative of F_{bd} to vanish at 0 and 1, and the compact support of the smooth up forces all its derivatives to vanish at ± 1. □

In particular $F'_{\mathrm{bd}}(x) = 2F_{\mathrm{bd}}(2x)$ on $[0, \frac{1}{2}]$. This differential equation must not be extended to all real x with the *bounded* F_{bd} on the right-hand side; the signed extension of the next subsection is the function that satisfies it globally.

2.2 The signed extension and the dyadic jet

Define the locally finite sum

$$F_{\mathrm{gl}}(x) := \sum_{b=0}^{\infty} \varepsilon_b \, \mathrm{up}(x - 2b - 1). \tag{20}$$

On $[2b, 2b+2]$ only the bth term can be nonzero, so

$$F_{\mathrm{gl}}(x) = \varepsilon_b \, \mathrm{up}(x - 2b - 1) \quad (2b \le x \le 2b+2), \qquad F_{\mathrm{gl}} = F_{\mathrm{bd}} \text{ on } [0,1], \qquad F_{\mathrm{gl}} = 0 \text{ on } (-\infty, 0]. \tag{21}$$

At the shared endpoints $x = 2b$ both adjacent formulas give zero, so the choice of block is immaterial.

Lemma 2.2 (Dilation and dyadic termination)**.** *The function F_{gl} is C^∞ on $\mathbb{R}$ and satisfies*

$$F'_{\mathrm{gl}}(x) = 2F_{\mathrm{gl}}(2x), \qquad F^{(r)}_{\mathrm{gl}}(x) = 2^{T_r} F_{\mathrm{gl}}(2^r x) \quad (r \ge 0). \tag{22}$$

At nonnegative integers and half-integers,

$$F_{\mathrm{gl}}(2b) = 0, \qquad F_{\mathrm{gl}}(2b+1) = \varepsilon_b, \qquad F_{\mathrm{gl}}\left(\frac{m}{2}\right) = \frac{1}{2}\,\varepsilon_{\lfloor m/4 \rfloor} \quad (m \text{ odd}). \tag{23}$$

At a dyadic $x = m/2^n \in [0,1]$,

$$F^{(r)}_{\mathrm{bd}}(x) = 0 \qquad (r \ge n+1), \tag{24}$$

and likewise $\mathrm{up}^{(r)}(x) = 0$ for $r \ge n+1$ when $|x| \le 1$ is dyadic of depth n.

Proof. Flatness of up at ± 1 makes the locally finite sum smooth, including at the joining points. Insert Equation (17) into Equation (20):

$$F'_{\mathrm{gl}}(x) = 2\sum_{b \ge 0} \varepsilon_b \big\{ \mathrm{up}(2x - 4b - 1) - \mathrm{up}(2x - 4b - 3) \big\} = 2 \sum_{c \ge 0} \varepsilon_c \, \mathrm{up}(2x - 2c - 1) = 2F_{\mathrm{gl}}(2x),$$

because the digit definition gives $\varepsilon_{2b} = \varepsilon_b$ and $\varepsilon_{2b+1} = -\varepsilon_b$, so the two terms are the even and odd summands of the single sum. Repeated differentiation and the identity $T_r + r = T_{r+1}$ prove Equation (22). The integer values follow from the support of up and $\mathrm{up}(0) = 1$, $\mathrm{up}(\pm 1) = 0$; at an odd half-integer $m/2$ the argument of the relevant translate is $\pm\frac{1}{2}$, where $\mathrm{up} = F_{\mathrm{bd}}(\frac{1}{2}) = \frac{1}{2}$ by Equations (15) and (16), and the block index is $\lfloor m/4 \rfloor$.

If $r \ge n+1$, the argument $2^r m/2^n$ is an even integer, so Equation (22) gives zero. Derivatives of F_{bd} and F_{gl} agree at interior points of $[0,1]$ and, by smoothness and one-sided agreement, at the endpoints. On $(-1,1)$ one has $\mathrm{up}(x) = F_{\mathrm{gl}}(x+1)$, and the same reasoning applies to $2^r(x+1)$; at $x = \pm 1$ all derivatives of up vanish. □

Remark 2.3 (A terminating jet is not a local polynomial identity). The vanishing Equation (24) does not assert that F_{bd} equals its Taylor polynomial near a dyadic point — the Fabius function is nowhere analytic. The exact spline statements proved in Section 7 concern a *finite convolution* and a cell whose width shrinks with its depth, together with an exact smoothing identity; at no point is F_{bd} replaced by a convergent Taylor series. This distinction is essential to every proof below.

2.3 Moment generating functions and independent tails

Put

$$H(t) := \frac{\sinh t}{t}, \quad H(0) = 1, \qquad M(t) := \mathbb{E}\, \mathrm{e}^{tZ} = \prod_{j=1}^{\infty} H\left(\frac{t}{2^j}\right). \tag{25}$$

The product converges locally uniformly, since $H(t/2^j) = 1 + \mathcal{O}_K\big(4^{-j}\big)$ on each compact set K; it defines an entire even function, which is also the moment generating function of the bounded variable Z directly. Its even Taylor coefficients are the normalized moments

$$a_k := \left[t^{2k}\right] M(t) = \frac{\mathbb{E}Z^{2k}}{(2k)!}, \qquad c_k := \mathbb{E}Z^{2k} = (2k)!\, a_k, \tag{26}$$

and the odd moments vanish. Define the finite prefixes

$$Z_N := \sum_{j=1}^{N} 2^{-j} V_j, \qquad M_N(t) := \mathbb{E}\, \mathrm{e}^{tZ_N} = \prod_{j=1}^{N} H\!\left(\frac{t}{2^j}\right), \tag{27}$$

with $Z_0 = 0$, $M_0 = 1$, and the uncentred prefixes $X_N := \sum_{j=1}^{N} 2^{-j} U_j$. The exact factorizations are

$$M(t) = M_N(t)\, M(2^{-N} t), \qquad M(2t) = H(t)\, M(t), \tag{28}$$

the second being the first with $N = 1$ after the substitution $t \mapsto 2t$, or equivalently the statement $Z \overset{\text{law}}{=} (V + Z')/2$ on the transform side. Quotients by M will be used only near zero or as formal power series with constant term one; no assertion that $1/M$ is entire is needed. For the raw variable,

$$M_X(z) := \mathbb{E}\, \mathrm{e}^{zX} = \mathrm{e}^{z/2} M(z/2), \qquad A_m := \frac{\mathbb{E}X^m}{m!} = [z^m]\, M_X(z). \tag{29}$$

Comparing coefficients in $M_X(z) = \mathrm{e}^{z/2} M(z/2)$ gives the dictionary

$$A_m = 2^{-m} \sum_{k=0}^{\lfloor m/2 \rfloor} \frac{a_k}{(m-2k)!}, \tag{30}$$

so a finite formula for the a_k is also one for the A_m, and conversely.

3 From a finite box to the dyadic master identity

3.1 A finite-convolution formula with proof

Lemma 3.1 (Box–simplex inclusion–exclusion). *Let $w_1, \dots, w_N > 0$ and let $U_1, \dots, U_N$ be independent uniforms on $[0, 1]$. For $N \geq 1$ and every real x,*

$$\mathbb{P}\Big(\sum_{j=1}^{N} w_j U_j \leq x\Big) = \frac{1}{N! \prod_{j=1}^{N} w_j} \sum_{\delta \in \{0,1\}^N} (-1)^{|\delta|} \left(x - \sum_{j=1}^{N} w_j \delta_j\right)_+^N, \tag{31}$$

where $|\delta| = \sum_j \delta_j$ and $s_+ = \max(s, 0)$.

Proof. After the substitution $y_j = w_j U_j$, the probability is the volume of the part of the box $\prod_j [0, w_j]$ on which $\sum_j y_j \leq x$, divided by $\prod_j w_j$. The unrestricted nonnegative simplex $\Big\{y \geq 0 : \sum_j y_j \leq x\Big\}$ has volume $x_+^N/N!$, by scaling the unit simplex, whose volume is $1/N!$ by N successive integrations. Exclude the conditions $y_j > w_j$ by inclusion–exclusion: for a fixed set $S = \{j : \delta_j = 1\}$ of violated constraints, translate y_j by w_j for $j \in S$; the remaining region is again a nonnegative simplex, with threshold $x - \sum_{j \in S} w_j$. Summing its volume with the sign $(-1)^{|S|}$ gives Equation (31). Boundaries have zero volume, so the weak inequalities cause no ambiguity. $\square$

For $w_j = 2^{-j}$ the subset sums are $h/2^N$ with $0 \le h < 2^N$: reading the digits of h from the most significant end, $h = \sum_{j=1}^{N} \delta_j 2^{N-j}$, and the sign $(-1)^{|\delta|}$ is the Thue–Morse sign ε_h. The product of the weights is 2^{-T_N}. Hence the distribution function H_N of the uncentred prefix X_N is

$$H_N(x) = \frac{2^{T_N}}{N!} \sum_{h=0}^{2^N-1} \varepsilon_h \left(x - \frac{h}{2^N}\right)_+^N. \tag{32}$$

This is a finite formula for an approximant, not yet a finite formula for F_{bd}.

3.2 Why the tail becomes polynomial at a dyadic point

The independent-copy decomposition $X \overset{\text{law}}{=} X_n + 2^{-n}X'$, with X' a copy of X independent of X_n, gives $F_{\mathrm{bd}}(x) = \mathbb{E}H_n(x - 2^{-n}X')$ for every real x. At $x = m/2^n$ the active set in Equation (32) is particularly simple: the argument $m - h - X'$ is nonnegative for every $X' \in [0,1]$ when $h < m$ and nonpositive when $h \ge m$. The support $0 \le X' \le 1$ of the tail is exactly what makes this possible.

Theorem 3.2 (Dyadic master identity). *For $n \ge 1$ and $0 \le m \le 2^n$,*

$$F_{\mathrm{bd}}\left(\frac{m}{2^n}\right) = 2^{-\binom{n}{2}} \sum_{h=0}^{m-1} \varepsilon_h \sum_{l=0}^{n} A_l \frac{(m-h-1)^{n-l}}{(n-l)!} \tag{33}$$

$$= 2^{-T_n} \sum_{h=0}^{m-1} \varepsilon_h \sum_{k=0}^{\lfloor n/2 \rfloor} a_k \frac{(2m-2h-1)^{n-2k}}{(n-2k)!} \tag{34}$$

$$= \frac{2^{-T_n}}{n!} \sum_{h=0}^{m-1} \varepsilon_h \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} c_k \,(2m-2h-1)^{n-2k}. \tag{35}$$

The centred form also holds for $n = 0$ with the empty sum and $a_0 = 1$, giving $F_{\mathrm{bd}}(0) = 0$ and $F_{\mathrm{bd}}(1) = 1$.

Proof. Substitute $x = m/2^n$ in $F_{\mathrm{bd}}(x) = \mathbb{E}H_n(x - 2^{-n}X')$ and use Equation (32) at $N = n$:

$$F_{\mathrm{bd}}\left(\frac{m}{2^n}\right) = \frac{2^{T_n}}{n!} \sum_{h=0}^{2^n-1} \varepsilon_h \,\mathbb{E}\left(\frac{m-h-X'}{2^n}\right)_+^n = \frac{2^{T_n-n^2}}{n!} \sum_{h=0}^{m-1} \varepsilon_h \,\mathbb{E}(m-h-X')^n,$$

where the inactive terms $h \ge m$ were dropped and the positive part removed for $h < m$; note $T_n - n^2 = -\binom{n}{2}$. By Equation (16), $X' \overset{\text{law}}{=} 1 - X'$, so the expectation equals $\mathbb{E}(m-h-1+X)^n$; expanding the binomial and using $\mathbb{E}X^l/l! = A_l$ gives Equation (33). Next write $m - h - 1 + X = \frac{1}{2}(2m - 2h - 1 + Z)$: the extra factor 2^{-n} turns $2^{-\binom{n}{2}}$ into 2^{-T_n}, the odd moments of Z vanish, and $\binom{n}{2k}c_k/n! = a_k/(n-2k)!$ gives Equation (34) and Equation (35). Symmetry of Z permits either sign in front of it. For $n = 0$ the centred display reads $\sum_{h<m} \varepsilon_h$ with $m \in \{0,1\}$. $\square$

This proof explains why the tail produces only finitely many moments: the dyadic threshold falls between consecutive head knots, so the truncated power is an ordinary polynomial throughout the tail's support. Rationality is not assumed anywhere.

Remark 3.3 (The Taylor-integral proof). The signed extension gives a second proof that also covers numerators larger than 2^n. Taylor's formula with integral remainder at the flat point 0, of order $n+1$, and the dilation law Equation (22) give

$$F_{\mathrm{gl}}(x) = \frac{1}{n!} \int_0^x (x-t)^n F_{\mathrm{gl}}^{(n+1)}(t)\,\mathrm{d}t = \frac{2^{-T_n}}{n!} \int_0^{2^{n+1}x} (2^{n+1}x - v)^n F_{\mathrm{gl}}(v)\,\mathrm{d}v$$

after the substitution $v = 2^{n+1}t$. At $x = a/2^n$ the integration interval is $[0, 2a]$, a union of a complete signed bump blocks; on the hth block substitute $v = 2h + 1 + y$ and use Equation (21):

$$F_{\mathrm{gl}}\left(\frac{a}{2^n}\right) = \frac{2^{-T_n}}{n!}\sum_{h=0}^{a-1}\varepsilon_h\int_{-1}^{1}(2a-2h-1-y)^n\,\mathrm{up}(y)\;\mathrm{d}y = 2^{-T_n}\sum_{h=0}^{a-1}\varepsilon_h\sum_{k=0}^{\lfloor n/2\rfloor}a_k\frac{(2a-2h-1)^{n-2k}}{(n-2k)!}, \tag{36}$$

valid for every $a \geq 0$ and $n \geq 0$. For $0 \leq a \leq 2^n$ the left side is $F_{\mathrm{bd}}(a/2^n)$ and the display is Equation (34).

For later use set

$$\Phi_r(v) := [t^r]\;\mathrm{e}^{vt}M(t) = \sum_{k=0}^{\lfloor r/2\rfloor}a_k\,\frac{v^{r-2k}}{(r-2k)!}, \qquad \Phi_0 = 1, \tag{37}$$

so that Equation (34) reads

$$F_{\mathrm{bd}}\left(\frac{m}{2^n}\right) = 2^{-T_n}\sum_{h=0}^{m-1}\varepsilon_h\,\Phi_n(2m-2h-1) = 2^{-T_n}\;[t^n]\;M(t)\sum_{h=0}^{m-1}\varepsilon_h\,\mathrm{e}^{(2m-2h-1)t}. \tag{38}$$

At this stage the moments have merely been isolated. The next section replaces every a_k, and every A_l, by a finite expression.

4 Eliminating every moment

4.1 Bernoulli numbers without a recurrence

Define, for every nonnegative integer s,

$$\beta_s := \sum_{j=0}^{s}\frac{1}{j+1}\sum_{v=0}^{j}(-1)^v\binom{j}{v}v^s, \tag{39}$$

with $0^0 = 1$.

Lemma 4.1 (Finite Bernoulli identity)**.** *As formal power series, $\sum_{s\geq 0}\beta_s z^s/s! = z/(\mathrm{e}^z-1)$. Consequently $\beta_s = B_s$ with $B_1 = -\frac{1}{2}$, and $B_{2r+1} = 0$ for $r \geq 1$.*

Proof. The exponential generating function of the inner finite difference is $\sum_s \frac{z^s}{s!}\sum_{v=0}^{j}(-1)^v\binom{j}{v}v^s = (1-\mathrm{e}^z)^j$, which has order j in z; hence the inner sum vanishes for $j > s$, and the sum over j in Equation (39) may be extended to infinity without changing any coefficient. With $u = 1 - \mathrm{e}^z$,

$$\sum_{j\geq 0}\frac{u^j}{j+1} = \frac{-\log(1-u)}{u} = \frac{z}{\mathrm{e}^z-1}.$$

All operations are formal and locally finite in each degree. Finally $z/(\mathrm{e}^z-1)+z/2 = \frac{z}{2}\coth\frac{z}{2}$ is even, which gives the odd-index vanishing. ☐

4.2 Explicit cumulants

For $r \geq 1$ put

$$\lambda_r := \frac{2^{2r-1}B_{2r}}{r\,(2r)!\,(4^r-1)}, \qquad \beta_r^* := \frac{\lambda_r}{4^r} = \frac{B_{2r}}{2r\,(2r)!\,(4^r-1)}. \tag{40}$$

The Bernoulli numbers on the right can always be replaced by Equation (39).

Proposition 4.2 (Explicit cumulants). *Near zero, and also formally,*

$$\log M(t) = \sum_{r\ge 1} \lambda_r t^{2r}, \qquad \log M_X(z) = \frac{z}{2} + \sum_{r\ge 1} \beta_r^* z^{2r}. \tag{41}$$

Thus the even cumulant of Z of order $2r$ is $(2r)!\lambda_r$, every odd cumulant of Z vanishes, and the cumulants of X beyond the mean $\frac{1}{2}$ are $(2r)!\beta_r^*$. The first values are

$$\lambda_1 = \frac{1}{18}, \quad \lambda_2 = -\frac{1}{2700}, \quad \lambda_3 = \frac{1}{178605}, \quad \lambda_4 = -\frac{1}{9639000}, \quad \lambda_5 = \frac{1}{478533825}. \tag{42}$$

Proof. Differentiate $\log H(t)$:

$$\frac{\mathrm{d}}{\mathrm{d}t} \log H(t) = \coth t - \frac{1}{t} = 1 + \frac{2}{\mathrm{e}^{2t}-1} - \frac{1}{t} = \frac{1}{t}\Big(\frac{2t}{\mathrm{e}^{2t}-1} + t - 1\Big) = \sum_{r\ge 1} \frac{2^{2r} B_{2r}}{(2r)!} t^{2r-1},$$

by Theorem 4.1 at $z = 2t$: the constant term cancels because $B_1 = -\frac{1}{2}$, and the odd Bernoulli numbers vanish. Integrating with zero constant term gives $\log H(t) = \sum_{r\ge 1} \frac{2^{2r-1} B_{2r}}{r(2r)!} t^{2r}$. Summing this at $t/2^j$ over $j \ge 1$ multiplies the coefficient of t^{2r} by $\sum_{j\ge 1} 4^{-rj} = 1/(4^r - 1)$; local uniform convergence of the product justifies the analytic argument, and coefficientwise geometric summation gives the same formal result. The raw statement follows from $M_X(z) = \mathrm{e}^{z/2} M(z/2)$ and $\lambda_r 4^{-r} = \beta_r^*$. □

Bernoulli numbers need not be accepted even as finite sums: the cumulant coefficients also have a formula using only factorials.

Lemma 4.3 (Bernoulli-free cumulant formula). *For every $r \ge 1$,*

$$\boxed{\lambda_r = \frac{1}{4^r - 1} \sum_{\ell=1}^{r} \frac{(-1)^{\ell-1}}{\ell} \sum_{\substack{s_1,\dots,s_\ell \ge 1 \\ s_1+\dots+s_\ell = r}} \prod_{i=1}^{\ell} \frac{1}{(2s_i+1)!}\,.} \tag{43}$$

Proof. Write $H(t) = 1 + u(t)$ with $u(t) = \sum_{s\ge 1} t^{2s}/(2s+1)!$, a formal series without constant term. Then $\log H(t) = \sum_{\ell \ge 1} \frac{(-1)^{\ell-1}}{\ell} u(t)^\ell$, and the coefficient of t^{2r} in u^ℓ is the sum over ordered ℓ-tuples of positive integers with sum r of $\prod_i 1/(2s_i+1)!$; only $\ell \le r$ contributes. Hence $\left[t^{2r}\right] \log H(t)$ is the double sum in Equation (43) without the prefactor. Summing over the scales $t/2^j$, $j \ge 1$, as in the previous proof multiplies it by $1/(4^r - 1)$. Comparison with Equation (41) gives the claim (and, incidentally, a factorial-only formula for $2^{2r-1} B_{2r}/(r(2r)!)$). □

4.3 Multiplicity-profile coefficients

Theorem 4.4 (Partition formulas for the moment coefficients). *For every $k, m \ge 0$,*

$$a_k = \sum_{(v_1,\dots,v_k) \in \mathcal{P}_k} \prod_{r=1}^{k} \frac{\lambda_r^{v_r}}{v_r!}, \tag{44}$$

$$A_m = \sum_{\substack{v_0, v_1, \dots, v_{\lfloor m/2 \rfloor} \ge 0 \\ v_0 + 2\sum_{r\ge 1} r v_r = m}} \frac{2^{-v_0}}{v_0!} \prod_{r=1}^{\lfloor m/2 \rfloor} \frac{(\beta_r^*)^{v_r}}{v_r!}. \tag{45}$$

At index zero each sum consists of the empty or zero vector and has value one.

Proof. Exponentiating Equation (41) gives $M(t) = \prod_{r\ge 1} \exp(\lambda_r t^{2r})$. To extract the coefficient of t^{2k} only the factors with $r \le k$ and the powers $v_r \le k/r$ can contribute; expanding these finitely many exponentials to the relevant orders and choosing v_r copies of $\lambda_r t^{2r}$ gives exactly Equation (44). The second identity follows in the same way from $M_X(z) = \mathrm{e}^{z/2}\prod_r \exp(\beta_r^* z^{2r})$, the factor $\mathrm{e}^{z/2}$ contributing v_0 copies of $z/2$. The same calculation is valid coefficientwise in formal power series, so no analytic evaluation is required. These are the multiplicity-vector versions of the complete Bell-polynomial identities, written out so that no convention for Bell polynomials is needed. □

Theorem 4.5 (Fully expanded Bernoulli partition value)**.** *Let $n \ge 0$, $0 \le m \le 2^n$, and $d = \lfloor n/2 \rfloor$. Then*

$$\boxed{F_{\mathrm{bd}}\Big(\frac{m}{2^n}\Big) = 2^{-T_n} \sum_{h=0}^{m-1} \varepsilon_h \sum_{\substack{v_0, v_1, \dots, v_d \ge 0 \\ v_0 + 2\sum_{r=1}^{d} r v_r = n}} \frac{(2m-2h-1)^{v_0}}{v_0!} \prod_{r=1}^{d} \frac{\lambda_r^{v_r}}{v_r!}\,,} \tag{46}$$

where every λ_r is the rational expression Equation (40), with B_{2r} given by Equation (39) or λ_r by Equation (43).

Proof. Substitute Equation (44) into Equation (34) and set $v_0 = n - 2k$; conversely every vector in Equation (46) determines $k = \sum_r r v_r$, so the reindexing is a bijection. Equivalently, expand $\mathrm{e}^{vt}\exp(\sum_r \lambda_r t^{2r})$ in Equation (38). □

4.4 Positive ordered compositions: no Bernoulli numbers

Theorem 4.6 (Positive composition formulas)**.** *For every $k, m \ge 0$, with the suffix sums Equation (9),*

$$a_k = \sum_{\mathbf{r}\in\mathcal{C}_k} \prod_{i=1}^{\ell(\mathbf{r})} \frac{1}{(2r_i+1)!\,(4^{R_i}-1)}, \tag{47}$$

$$A_m = \sum_{\mathbf{r}\in\mathcal{C}_m} \prod_{i=1}^{\ell(\mathbf{r})} \frac{1}{(r_i+1)!\,(2^{R_i}-1)}. \tag{48}$$

Every summand is positive. Together with $a_0 = A_0 = 1$ this supplies every coefficient in Theorem 3.2 without any special-number sequence.

Proof. Expand the product Equation (25) using $H(t/2^j) = \sum_{r\ge 0} 4^{-jr} t^{2r}/(2r+1)!$. In a contribution to degree $2k$, let $j_1 < \cdots < j_\ell$ be the distinct occupied scales and $r_i \ge 1$ the exponent chosen at scale j_i, so $(r_1, \dots, r_\ell) \in \mathcal{C}_k$. Its weight is $\prod_i 4^{-j_i r_i}/(2r_i+1)!$. All coefficients are nonnegative, so the terms may be regrouped by the composition (monotone convergence, or first do the argument for the finite product M_N and let $N \to \infty$, the even coefficients increasing to those of M). Put $g_1 = j_1$ and $g_i = j_i - j_{i-1}$ for $i > 1$: the gaps are positive integers in bijection with the increasing tuples, and $\sum_i j_i r_i = \sum_i g_i R_i$, since $j_i = g_1 + \cdots + g_i$ and each g_i multiplies exactly the exponents $r_i, \dots, r_\ell$. Hence the sum over occupied scales factors into finite geometric sums,

$$\sum_{1\le j_1<\cdots<j_\ell} 4^{-\sum_i j_i r_i} = \prod_{i=1}^{\ell} \sum_{g_i \ge 1} 4^{-g_i R_i} = \prod_{i=1}^{\ell} \frac{1}{4^{R_i}-1}, \tag{49}$$

which proves Equation (47). For the raw coefficients expand $M_X(z) = \prod_{j\ge 1} \mathbb{E}\,\mathrm{e}^{2^{-j} z U_j}$ with $\mathbb{E}U^r/r! = 1/(r+1)!$; the gap calculation is identical with 2 in place of 4, giving Equation (48). □

Example 4.7 (First coefficients)**.** The compositions (1), and (2), $(1,1)$, give

$$a_1 = \frac{1}{3!\,(4-1)} = \frac{1}{18}, \qquad a_2 = \frac{1}{5!\,(16-1)} + \frac{1}{3!\,3!\,(16-1)(4-1)} = \frac{19}{16200},$$

so $\mathbb{E}Z^2 = \frac{1}{9}$ and $\mathbb{E}Z^4 = \frac{19}{675}$; the four compositions of 3 give $a_3 = 583/42865200$, and $a_4 = 132809/1311675120000$. The same numbers result from Equation (44): $a_2 = \lambda_2 + \lambda_1^2/2$ and $a_3 = \lambda_3 + \lambda_1\lambda_2 + \lambda_1^3/6$. Raw: $A_1 = \frac{1}{2}$, $A_2 = \frac{5}{36}$, $A_3 = \frac{1}{36}$, $A_4 = \frac{143}{32400}$. Multiplying normalized coefficients by $(2k)!$ is essential; omitting the factor would change the dyadic values.

Theorem 4.8 (Bernoulli-free closed dyadic sum). *For $n \geq 0$, $0 \leq m \leq 2^n$, $d = \lfloor n/2 \rfloor$,*

$$\boxed{\begin{aligned} F_{\mathrm{bd}}\left(\frac{m}{2^n}\right) &= 2^{-T_n} \sum_{h=0}^{m-1} \varepsilon_h \sum_{k=0}^{d} \frac{(2m-2h-1)^{n-2k}}{(n-2k)!} \\ &\quad \times \sum_{\ell=0}^{k} \sum_{\substack{r_1,\dots,r_\ell \geq 1 \\ r_1+\cdots+r_\ell = k}} \prod_{i=1}^{\ell} \frac{1}{(2r_i+1)!\,\left(4^{r_i+\cdots+r_\ell}-1\right)}. \end{aligned}} \tag{50}$$

The $\ell = 0$ inner sum is one for $k = 0$ and zero otherwise. Every quantity is a factorial, an integer power, a Thue–Morse sign, or a bounded sum or product of these.

Proof. Insert Equation (47) into Equation (34). □

Remark 4.9 (Relation to the triangular recurrence). Comparing coefficients of t^{2k} in $M(2t) = H(t)M(t)$ gives

$$(4^k - 1)\, a_k = \sum_{j=0}^{k-1} \frac{a_j}{(2(k-j)+1)!} \qquad (k \geq 1). \tag{51}$$

Unrolling this recurrence produces a path interpretation with the same suffix-composition weights as Equation (47), so the composition formula is the closed form of the recurrence, not a recurrence in disguise. Equation Equation (51) is retained as an independent check and as the source of the matrix in Section 10; it is not part of any evaluation prescription.

The composition sum has 2^{k-1} terms for $k \geq 1$, all positive; the partition sum has one term per integer partition of k, at the price of alternating signs among the λ_r. Both are exact rational formulas, and neither requires $a_0, \dots, a_{k-1}$ before a_k.

4.5 Finite reciprocal coefficients

The scale-polynomial results of Section 7 use

$$\frac{1}{M(t)} = \sum_{k \geq 0} e_k t^{2k}, \qquad e_0 = 1, \tag{52}$$

as a formal series or near zero. Exactly the calculation of Theorem 4.4 applied to $\exp(-\sum_r \lambda_r t^{2r})$ gives the recurrence-free formula

$$e_k = \sum_{(v_1,\dots,v_k) \in \mathcal{P}_k} \prod_{r=1}^{k} \frac{(-\lambda_r)^{v_r}}{v_r!}, \tag{53}$$

with first values

$$e_1 = -\frac{1}{18}, \quad e_2 = \frac{31}{16200}, \quad e_3 = -\frac{2347}{42865200}, \quad e_4 = \frac{1904369}{1311675120000}. \tag{54}$$

Alternatively, the λ_r can be avoided by the finite geometric expansion of $1/(1 + (M-1))$:

$$e_k = \sum_{\ell=1}^{k} (-1)^\ell \sum_{\substack{r_1,\dots,r_\ell \geq 1 \\ r_1+\cdots+r_\ell = k}} \prod_{i=1}^{\ell} a_{r_i} \qquad (k \geq 1), \tag{55}$$

followed by Equation (47). The signs $(-1)^k e_k > 0$ are proved in Theorem 7.6.

4.6 Reciprocal powers of two

At $m = 1$ the raw form Equation (33) collapses to a single coefficient:

$$\boxed{F_{\mathrm{bd}}(2^{-n}) = 2^{-\binom{n}{2}} A_n} \qquad (n \geq 0), \tag{56}$$

so either Equation (45) or Equation (48) is a direct finite formula for these values, for example $F_{\mathrm{bd}}(\frac{1}{16}) = 2^{-6} A_4 = 143/2073600$. In centred form, Equation (34) at $m = 1$ gives the all-positive expression

$$F_{\mathrm{bd}}(2^{-n}) = 2^{-T_n} \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{1}{(n-2k)!} \sum_{\mathbf{r} \in \mathcal{C}_k} \prod_{i=1}^{\ell(\mathbf{r})} \frac{1}{(2r_i+1)!\,(4^{R_i}-1)}. \tag{57}$$

For odd depths there is a further simplification through the bump's half moments:

$$F_{\mathrm{bd}}\left(2^{-(2j+1)}\right) = \frac{c_j}{2\,(2j)!\,2^{\binom{2j+1}{2}}} = \frac{a_j}{2^{\binom{2j+1}{2}+1}} \qquad (j \geq 0). \tag{58}$$

To prove it without a moment recurrence, note that $\mathrm{up}(t) = F_{\mathrm{bd}}(1-t) = \mathbb{P}(X \geq t)$ on $[0,1]$ by Equations (15) and (16). Layer-cake integration gives $\mathbb{E}X^{2j+1} = (2j+1)\int_0^1 t^{2j}\mathbb{P}(X \geq t)\,\mathrm{d}t = (2j+1)\int_0^1 t^{2j}\,\mathrm{up}(t)\,\mathrm{d}t = (2j+1)\,c_j/2$, by evenness of up and $\int_{-1}^1 t^{2j}\,\mathrm{up} = c_j$; hence $A_{2j+1} = c_j/(2\,(2j)!)$, and Equation (56) gives the claim. For example $F_{\mathrm{bd}}(\frac{1}{8}) = a_1/2^4 = 1/288$ and $F_{\mathrm{bd}}(\frac{1}{32}) = a_2/2^{11} = 19/33177600$.

5 Binary compression of the numerator

The master formula sums over m consecutive integers. Its special signs permit an exact compression into binary blocks, in two different ways, and both are useful independently of which coefficient formula is chosen.

5.1 One outer summand per nonzero binary digit

Write the binary expansion and the digit data

$$m = \sum_{j \geq 0} b_j 2^j, \quad b_j \in \{0,1\}, \qquad J(m) := \{j : b_j = 1\}, \tag{59}$$

and for $j \in J(m)$

$$p_j := 2^{j+1}\left\lfloor \frac{m}{2^{j+1}} \right\rfloor, \qquad r_j := m \bmod 2^j, \qquad K_j := 2^j + 2r_j, \qquad \eta_j := \varepsilon_{\lfloor m/2^{j+1} \rfloor}. \tag{60}$$

Thus p_j is the part of m strictly above the jth bit, r_j the part strictly below it, and $m = p_j + 2^j + r_j$; these are digit operations, not recursive values of F_{bd}.

Theorem 5.1 (Binary block formula). *For every $a \geq 0$ and $n \geq 0$, including numerators larger than 2^n,*

$$\boxed{F_{\mathrm{gl}}\left(\frac{a}{2^n}\right) = 2^{-T_n} \sum_{\substack{j \in J(a) \\ j \leq n}} \eta_j\, 2^{T_j} \sum_{k=0}^{\lfloor (n-j)/2 \rfloor} \frac{a_k\, 4^{jk}\, K_j^{\,n-j-2k}}{(n-j-2k)!}}, \tag{61}$$

equivalently, with Φ_r from Equation (37) and $A_j := K_j/2^j = 1 + 2^{1-j} r_j$,

$$F_{\mathrm{gl}}\left(\frac{a}{2^n}\right) = \sum_{\substack{j \in J(a) \\ j \leq n}} \eta_j\, 2^{-T_{n-j}}\, \Phi_{n-j}(A_j). \tag{62}$$

For $0 \le a \le 2^n$ the left side is $F_{\mathrm{bd}}(a/2^n)$. The coefficients a_k may be supplied by Equation (47) or Equation (44); the outer sum has at most $\min(n+1, \mathrm{wt}_2(a))$ nonzero terms, independently of the size of a.

Proof. The interval of integers $[0,a)$ is the disjoint union of the blocks $[p_j, p_j+2^j)$, $j \in J(a)$. Adding an integer $h < 2^j$ to p_j causes no binary carry, so $\varepsilon_{p_j+h} = \varepsilon_{p_j}\varepsilon_h = \eta_j\varepsilon_h$. The finite Thue–Morse product identity

$$\sum_{h=0}^{2^j-1} \varepsilon_h\, z^h = \prod_{\ell=0}^{j-1}\left(1 - z^{2^\ell}\right) \tag{63}$$

holds because selecting a subset of the factors selects the nonzero digits of h. Therefore

$$\sum_{h=0}^{a-1} \varepsilon_h\, z^h = \sum_{j\in J(a)} \eta_j\, z^{p_j} \prod_{\ell=0}^{j-1}\left(1 - z^{2^\ell}\right). \tag{64}$$

By Equation (36), the desired value is $2^{-T_n}\,[t^n]\ \mathrm{e}^{(2a-1)t} M(t) \sum_{h<a} \varepsilon_h \mathrm{e}^{-2ht}$. For the block of length 2^j, each factor of the product at $z = \mathrm{e}^{-2t}$ is $1 - \mathrm{e}^{-2^{\ell+1}t} = 2^{\ell+1} t\, \mathrm{e}^{-2^\ell t} H(2^\ell t)$, so

$$\prod_{\ell=0}^{j-1}\left(1 - \mathrm{e}^{-2^{\ell+1}t}\right) = 2^{T_j}\, t^j\, \mathrm{e}^{-(2^j-1)t} \prod_{\ell=0}^{j-1} H(2^\ell t), \tag{65}$$

the powers of two multiplying to $2^{j+j(j-1)/2} = 2^{T_j}$. Iterating $M(2t) = H(t)M(t)$ gives the exact cancellation $M(t)\prod_{\ell=0}^{j-1} H(2^\ell t) = M(2^j t)$. The block therefore contributes

$$\eta_j\, 2^{-T_n+T_j} \left[t^{n-j}\right]\ \mathrm{e}^{(2a-2p_j-2^j)t} M(2^j t),$$

and $2a - 2p_j - 2^j = 2^j + 2r_j = K_j$. Expanding $M(2^j t) = \sum_k a_k 4^{jk} t^{2k}$ and extracting t^{n-j} proves Equation (61); a block with $j > n$ carries the factor t^j and contributes nothing, which is why the high bits enter only through the signs η_j. Substituting $s = 2^j t$ changes the coefficient extraction by the factor $2^{j(n-j)}$, turns the exponent into $A_j s$, and $-T_n + T_j + j(n-j) = -T_{n-j}$ gives Equation (62). ☐

For $a = 0$ the empty sum gives zero; for $a = 2^n$ the single block gives $\Phi_0(1) = 1$; for $a = 1$ only the lowest bit occurs and Equation (62) reduces to $F_{\mathrm{bd}}(2^{-n}) = 2^{-T_n}\Phi_n(1)$, which is Equation (57). Combining Equation (61) with Equation (47) gives one literal expression with no unresolved coefficient symbol,

$$\boxed{\begin{aligned} F_{\mathrm{gl}}\left(\frac{a}{2^n}\right) &= \sum_{\substack{j\in J(a)\\ j\le n}} \eta_j\, 2^{T_j - T_n} \sum_{k=0}^{\lfloor (n-j)/2\rfloor} \frac{4^{jk} K_j^{\,n-j-2k}}{(n-j-2k)!} \\ &\quad \times \sum_{\ell=0}^{k} \sum_{\substack{r_1,\dots,r_\ell \ge 1\\ r_1+\cdots+r_\ell = k}} \prod_{i=1}^{\ell} \frac{1}{(2r_i+1)!\left(4^{r_i+\cdots+r_\ell} - 1\right)}, \end{aligned}} \tag{66}$$

which, with the digit definitions Equation (60), contains no recursively defined constant at all. After the universal coefficients have been computed there are at most $\sum_{j=0}^{n}(\lfloor (n-j)/2\rfloor + 1) = (d+1)(n-d+1)$ argument-dependent monomials — a count of rational operations, not of bit complexity.

Example 5.2 (Three nonzero bits)**.** For $x = 21/64$, $21 = (10101)_2$ and $J(21) = \{0, 2, 4\}$. The block data give

$$\begin{aligned} F_{\mathrm{bd}}\left(\frac{21}{64}\right) &= 2^{-3}\Phi_2\left(\frac{13}{8}\right) - 2^{-10}\Phi_4\left(\frac{3}{2}\right) + 2^{-21}\Phi_6(1) \\ &= \frac{1585}{9216} - \frac{71179}{265420800} + \frac{1153}{561842749440} = \frac{482385079229}{2809213747200}. \end{aligned}$$

For $a = 3$, $n = 3$ the two blocks $(\eta_0, K_0) = (-1, 1)$ and $(\eta_1, K_1) = (1, 4)$ give $F_{\mathrm{bd}}(\frac{3}{8}) = \frac{1}{64}\big[-(\frac{1}{6} + \frac{1}{18}) + 2(\frac{16}{2} + \frac{4}{18})\big] = \frac{73}{288}$. Without a prefix sum of length 173,

$$F_{\mathrm{bd}}\left(\frac{173}{4096}\right) = \frac{149058844759931929437232685997667298875 7}{29221473288789871398691657592526707097600000 0} \approx 5.101003747717089{\cdot}10^{-6}, \tag{67}$$

where only the five set bits of $173 = 128 + 32 + 8 + 4 + 1$ contribute.

5.2 The binary telescope

A second compression uses raw coefficients and the signed extension's antiperiodicity rather than the block product; it is a finite telescope of the exact Taylor reduction used in the classical arithmetic theory [5].

Lemma 5.3 (One binary step). *For an integer $b \ge 1$ and $0 \le t \le 2^{-b}$,*

$$F_{\mathrm{bd}}(2^{-b} + t) + F_{\mathrm{bd}}(t) = 2^{-\binom{b}{2}} \sum_{k=0}^{b} A_{b-k} \frac{(2^b t)^k}{k!}. \tag{68}$$

Proof. On $[0, 2]$, Equation (21) and $\varepsilon_1 = -1$ give $F_{\mathrm{gl}}(z + 2) = -F_{\mathrm{gl}}(z)$. By Equation (22) the $(b+1)$st derivative of the left side of Equation (68) is $2^{T_{b+1}}\big[F_{\mathrm{gl}}(2+2^{b+1}t) + F_{\mathrm{gl}}(2^{b+1}t)\big] = 0$ for $0 \le 2^{b+1}t \le 2$, so the left side is a polynomial of degree at most b on the stated interval. At $t = 0$ every derivative of $F_{\mathrm{bd}}(t)$ vanishes, while for $0 \le k \le b$

$$F_{\mathrm{bd}}^{(k)}(2^{-b}) = 2^{T_k} F_{\mathrm{bd}}\big(2^{-(b-k)}\big) = 2^{T_k - \binom{b-k}{2}} A_{b-k} = 2^{bk - \binom{b}{2}} A_{b-k}$$

by Equation (22) and Equation (56), using $T_k - \binom{b-k}{2} = bk - \binom{b}{2}$; the endpoint $k = b$ uses $F_{\mathrm{bd}}(1) = A_0 = 1$. The Taylor polynomial of degree b at $t = 0$ is therefore exactly the right side of Equation (68). ☐

Theorem 5.4 (Explicit binary telescope). *Let $0 < x < 1$ be dyadic, with nonzero binary positions $x = \sum_{i=1}^{\rho} 2^{-b_i}$, $1 \le b_1 < \cdots < b_\rho$, and the explicitly known binary suffixes*

$$s_i := \sum_{j=i+1}^{\rho} 2^{b_i - b_j}, \qquad 0 \le s_i < 1. \tag{69}$$

Then

$$\boxed{F_{\mathrm{bd}}(x) = \sum_{i=1}^{\rho} (-1)^{i-1}\, 2^{-\binom{b_i}{2}} \sum_{k=0}^{b_i} \frac{s_i^k}{k!} \sum_{\mathbf{r} \in \mathcal{C}_{b_i - k}} \prod_{j=1}^{\ell(\mathbf{r})} \frac{1}{(r_j + 1)!\,(2^{R_j} - 1)}\,.} \tag{70}$$

The inner composition sum is $A_{b_i - k}$ and may equally be replaced by Equation (45).

Proof. Write $x_i = \sum_{j=i}^{\rho} 2^{-b_j}$ and $x_{\rho+1} = 0$. Since $x_i = 2^{-b_i} + x_{i+1}$ and $2^{b_i} x_{i+1} = s_i < 1$, Theorem 5.3 gives $F_{\mathrm{bd}}(x_i) + F_{\mathrm{bd}}(x_{i+1}) = P_i(s_i)$ with P_i the polynomial on the right of Equation (68). Multiply by $(-1)^{i-1}$ and sum over i: all interior values cancel and $F_{\mathrm{bd}}(x_{\rho+1}) = F_{\mathrm{bd}}(0) = 0$. Substituting Equation (48) gives the display. This is a proof by a finite telescope, not an instruction to evaluate a recursive sequence. ☐

For example $3/8 = 2^{-2} + 2^{-3}$ gives $F_{\mathrm{bd}}(\frac{3}{8}) = 2^{-1}\big(A_2 + \frac{1}{2}A_1 + \frac{1}{8}A_0\big) - 2^{-3}A_3 = \frac{73}{288}$. A point with ρ nonzero bits uses $\sum_i (b_i + 1) \le \rho(n + 1)$ polynomial terms once the raw coefficients through index n are available.

Remark 5.5 (The two compressions compared). Both formulas have one outer term per set bit, but they are different identities: the block formula groups the Thue–Morse prefix into complete blocks, whose sign structure is absorbed by the exact product identity Equation (65), uses the centred coefficients a_k with the largest index $\lfloor (n-j)/2 \rfloor$, and holds for the signed extension at every numerator; the telescope alternates signs down the bits, uses the raw coefficients A_l with index up to b_i, and is confined to the unit interval. Reflection $F_{\mathrm{bd}}(x) = 1 - F_{\mathrm{bd}}(1-x)$ may reduce the number of useful bits in either. Neither count is a bit-complexity estimate: large rational numerators and the coefficient generation still have costs.

6 Eliminating moments by finitely many uniform prefixes

The coefficients a_k admit a third recurrence-free representation: instead of expanding the infinite product, interpolate a coefficient of its finite prefixes M_N. The same device turns the whole master identity into a formula with finitely many uniform coordinates.

6.1 Two explicit finite-prefix coefficients

Write

$$a_{k,N} := \left[t^{2k}\right] M_N(t), \qquad a_{0,0} = 1, \quad a_{k,0} = 0 \ (k > 0). \tag{71}$$

Expanding the N factors gives the finite multinomial formula

$$a_{k,N} = \sum_{\substack{r_1,\ldots,r_N \geq 0 \\ r_1+\cdots+r_N = k}} \prod_{i=1}^{N} \frac{4^{-ir_i}}{(2r_i+1)!}, \tag{72}$$

a finite sum because every $r_i \leq k$. There is also a single signed power sum. Since $H(t/2^j) = 2^{j-1}t^{-1}(\mathrm{e}^{t/2^j} - \mathrm{e}^{-t/2^j})$, expanding the finite product gives, with $L_N = 1 - 2^{-N}$,

$$M_N(t) = 2^{N(N-1)/2}\, t^{-N} \sum_{h=0}^{2^N-1} \varepsilon_h\, \mathrm{e}^{(L_N - 2^{1-N}h)t} : \tag{73}$$

the subset sums $\sum_{j\in S} 2^{-j}$ run through $h/2^N$ with sign ε_h (reversing the order of the N digits does not change the sign), and the constant is $\prod_{j=1}^{N} 2^{j-1} = 2^{N(N-1)/2}$. The apparent pole is removable, the sum vanishing to order N at $t = 0$ by Prouhet's cancellation Equation (90) below. Expanding the exponentials,

$$\boxed{a_{k,N} = \frac{2^{N(N-1)/2 - N(2k+N)}}{(2k+N)!} \sum_{h=0}^{2^N-1} \varepsilon_h\, (2^N - 1 - 2h)^{2k+N}\, .} \tag{74}$$

For instance $a_{1,1} = \frac{1}{24}$; the raw second moment $\mathbb{E}Z_1^2 = \frac{1}{12}$ is not $a_{1,1}$, and the normalization by $(2k)!$ remains essential.

6.2 Exact coefficient extraction at the quarter base

The geometric weights $w_{d,j}(q)$ are defined and proved in Section 8.2; here only the identity Equation (94) is used, which says that $\sum_{j=0}^{d} w_{d,j}(q)P(zq^j) = P(0)$ for every polynomial P of degree at most d and every $z \neq 0$.

Theorem 6.1 (Finite-prefix moment elimination). *For every $k \geq 0$ and every integer $N_0 \geq 0$,*

$$\boxed{a_k = \sum_{j=0}^{k} w_{k,j}\, a_{k,N_0+j}\, ,} \tag{75}$$

with $w_{k,j} = w_{k,j}(\frac{1}{4})$ the explicit rational weights Equation (5) and each $a_{k,N}$ given by Equation (72) or Equation (74). No infinite moment remains on the right. A common row of degree $d \ge k$ may replace the row of degree k.

Proof. From $M_N(t) = M(t)/M(2^{-N}t)$ and Equation (52),

$$a_{k,N} = \sum_{r=0}^{k} a_{k-r}\, e_r\, (4^{-N})^r. \tag{76}$$

For fixed k this is a polynomial of degree at most k in 4^{-N} with constant term a_k. Apply Equation (94) with $q = \frac{1}{4}$, $z = 4^{-N_0}$ and the nodes $4^{-(N_0+j)}$. The occurrence of a_{k-r} in this proof is not an auxiliary recurrence in the final expression: all sampled coefficients on the right of Equation (75) are supplied by independent finite formulas. □

At $N_0 = 0$ and $k > 0$ the $j = 0$ term vanishes, so only the prefixes $1, \dots, k$ are needed: $a_1 = \frac{4}{3}a_{1,1} = \frac{1}{18}$. Inserting Equations (74) and (75) into Equation (37) and then into Equation (34) or Equation (62) produces another family consisting entirely of finite Thue–Morse sums, in which the interpolation depth is controlled by the moment index k and the recovered coefficient is reusable at many dyadic arguments.

6.3 A moment-free multinomial formula

Applying the same interpolation to the whole master polynomial gives a formula with finitely many uniform coordinates. For $N \ge 0$ and any A, define

$$C_{n,N}(A) := \frac{1}{n!}\mathbb{E}(A + Z_N)^n = \sum_{\substack{r_0, r_1, \dots, r_N \ge 0 \\ r_0 + 2r_1 + \cdots + 2r_N = n}} \frac{A^{r_0}}{r_0!} \prod_{j=1}^{N} \frac{4^{-jr_j}}{(2r_j+1)!}, \tag{77}$$

so $C_{n,0}(A) = A^n/n!$ and every summation index is bounded by n.

Theorem 6.2 (Finite-cube closed form)**.** *For every $m, n \ge 0$ with $0 \le m \le 2^n$ and $d = \lfloor n/2 \rfloor$,*

$$\boxed{F_{\mathrm{bd}}\left(\frac{m}{2^n}\right) = 2^{-T_n} \sum_{h=0}^{m-1} \varepsilon_h \sum_{j=0}^{d} w_{d,j}\, C_{n,j}(2m - 2h - 1)\,,} \tag{78}$$

and no moment of an infinite distribution occurs in its evaluation.

Proof. With $C_n(A) := [t^n]\, \mathrm{e}^{At} M(t) = \Phi_n(A)$ and $C_{n,N}(A) = [t^n]\, \mathrm{e}^{At} M(t)/M(2^{-N}t)$, the reciprocal expansion gives $C_{n,N}(A) = \sum_{r=0}^{d} e_r 4^{-Nr} C_{n-2r}(A)$, a polynomial of degree at most d in 4^{-N} with constant term $\Phi_n(A)$. By Equation (94) at $z = 1$, $\Phi_n(A) = \sum_{j=0}^{d} w_{d,j} C_{n,j}(A)$; substitute this into Equation (38). □

The consecutive depths $0, 1, \dots, d$ are convenient, not compulsory: for distinct nonnegative $N_0, \dots, N_d$ and $z_j = 4^{-N_j}$, the Lagrange weights $\prod_{\ell \ne j}(-z_\ell)/(z_j - z_\ell)$ replace $w_{d,j}$ (Theorem 8.1 in the same form), and for $N_j = N + Lj$ they reduce to $w_{d,j}(4^{-L})$.

7 The exact polynomial hidden in centred finite splines

This section explains why a short moment-free formula exists. It is important that the cutoff dependence terminates *exactly*, not merely to a prescribed asymptotic accuracy.

7.1 The centred finite distribution

For $N \geq 1$ define

$$Y_N := X_N + 2^{-N-1} = \frac{1+Z_N}{2}, \qquad G_N(x) := \mathbb{P}(Y_N \leq x). \tag{79}$$

The added midpoint 2^{-N-1} is the mean of the omitted tail; this centring makes the correction polynomial even in the small scale, a simplification the uncentred H_N does not enjoy (Section 9). The finite random variable is centred at $\frac{1}{2}$, like X, and G_N is a piecewise polynomial of degree N whose knots lie among the points $(h + \frac{1}{2})/2^N$, $0 \leq h < 2^N$. The box formula gives

$$G_N(x) = \frac{2^{T_N}}{N!} \sum_{h=0}^{2^N-1} \varepsilon_h \left(x - \frac{h+\frac{1}{2}}{2^N} \right)_+^N . \tag{80}$$

At $x = m/2^n$ with $N \geq n \geq 1$, put $A = 2^{N-n}m$; exactly the terms $h < A$ are active, every one a full power, and

$$\boxed{G_N\left(\frac{m}{2^n}\right) = \frac{1}{2^{T_N} N!} \sum_{h=0}^{A-1} \varepsilon_h \, (2A - 2h - 1)^N = \frac{\mathcal{S}_N(A)}{2^{T_N} N!} \, .} \tag{81}$$

The threshold is always halfway between two consecutive possible knots. The sum is empty at $m = 0$ and evaluates to one at $m = 2^n$. Its negative terms come only from inclusion–exclusion; G_N itself is a distribution function.

7.2 A finite expression for every dyadic derivative

Lemma 7.1 (Differentiated master formula)**.** *For $n \geq 1$, $0 \leq m \leq 2^n$, and $0 \leq r \leq n$,*

$$F_{\mathrm{bd}}^{(r)}\left(\frac{m}{2^n}\right) = 2^{(n+1)r - T_n} \sum_{h=0}^{m-1} \varepsilon_h \, \Phi_{n-r}(2m - 2h - 1). \tag{82}$$

For $r \geq n+1$ the derivative is zero. In particular every derivative at a dyadic point is explicitly rational.

Proof. Conditioning on the independent tail gives, for every real y,

$$F_{\mathrm{bd}}(y) = \frac{2^{T_n}}{n!} \sum_{h=0}^{2^n-1} \varepsilon_h \, \mathbb{E}\left(y - 2^{-n}(h + X')\right)_+^n .$$

For $r \leq n$ differentiate r times under the expectation; when $r = n$ the derivative of the truncated power is $n!$ times the indicator of a positive argument, and X' has no atoms, so the value at the cutoff is immaterial. At $y = m/2^n$ all terms with $h < m$ have a nonnegative argument for every $X' \in [0,1]$ and the remaining terms vanish. Replacing X' by $1 - X' = (1-Z)/2$ and using evenness of Z, the scale is $2^{T_n} \cdot 2^{-n(n-r)} \cdot 2^{-(n-r)} = 2^{(n+1)r - T_n}$ and the expectation divided by $(n-r)!$ is Φ_{n-r}. Higher derivatives vanish by Theorem 2.2. □

Comparing with Equation (22) evaluated by the master identity, this is also the statement $F_{\mathrm{bd}}^{(r)}(x) = 2^{T_r} F_{\mathrm{gl}}(2^r x)$ written out; Section 11.4 returns to it.

7.3 Finite-jet deconvolution

Let D denote differentiation of polynomials in the variable s. The next lemma is the one analytic step; it is stated generally because it is used four times (for the CDF, the density, the shifted spline, and the integer bases).

Lemma 7.2 (Local polynomial smoothing is invertible on a finite jet)**.** *Let H be a piecewise polynomial that agrees with one polynomial $p(s)$ of degree at most N at the points $x+s$, $|s| \le \eta$, and let W be a random variable supported on $[-1,1]$ with a bounded density. Put $f(t) := \mathbb{E}H(t-\eta W)$ and let $M_W(z) = \mathbb{E}\,e^{zW}$. Suppose the derivatives of f through order N at x are obtained by convolving the corresponding piecewise derivatives of H. Then, as an identity of polynomials of degree at most N,*

$$\sum_{r=0}^{N} \frac{f^{(r)}(x)}{r!} s^r = M_W(-\eta \mathrm{D})\, p(s), \tag{83}$$

where the operator is truncated at order N, and consequently

$$p(s) = \big[M_W(-\eta \mathrm{D})\big]^{-1} \sum_{r=0}^{N} \frac{f^{(r)}(x)}{r!} s^r, \tag{84}$$

the inverse being the finite operator obtained by truncating the formal reciprocal series at order N. If W is symmetric, only even powers of $\eta\mathrm{D}$ occur. For the distribution-function splines of this section and the density splines of Section 11.2, the stipulated derivative identities hold.

Proof. For $|s| \le \eta$ and $|W| \le 1$ the point $x+s-\eta W$ may leave the cell, so the identity is proved at the level of jets. Differentiating the convolution as stipulated and evaluating at x, $f^{(r)}(x) = \mathbb{E}H^{(r)}(x-\eta W) = \mathbb{E}p^{(r)}(-\eta W)$ for $0 \le r \le N$, because $x-\eta W$ lies in the cell. Taylor's formula for the polynomial $p^{(r)}$ gives $p^{(r)}(-\eta W) = \sum_i p^{(r+i)}(0)(-\eta W)^i/i!$, and taking expectations coefficient by coefficient, $f^{(r)}(x) = \sum_i \mathbb{E}[W^i](-\eta)^i p^{(r+i)}(0)/i!$, which is the coefficient identity behind Equation (83). On polynomials of degree at most N one has $\mathrm{D}^{N+1} = 0$, so $M_W(-\eta\mathrm{D})$ is a unipotent finite operator ($M_W(0) = 1$), invertible with inverse the truncated reciprocal series; applying it gives Equation (84). No convergence of a Taylor series of f is assumed anywhere. For an N-uniform distribution function the derivatives through order $N-1$ are continuous and the Nth is bounded and piecewise constant; these are also its distributional derivatives, so convolution with a bounded density gives the derivative identities, values at the finitely many knots not affecting any integral. The density spline is the same argument one degree lower. □

7.4 The exact shrinking-cell identity

Let

$$P_x(h) := \sum_{r=0}^{n} \frac{F_{\mathrm{bd}}^{(r)}(x)}{r!} h^r, \qquad x = \frac{m}{2^n}, \tag{85}$$

the Taylor polynomial of F_{bd} at x, which terminates by Equation (24); no equality $F_{\mathrm{bd}}(x+h) = P_x(h)$ is asserted.

Theorem 7.3 (Exact shrinking-cell formula)**.** *Let $N \ge n \ge 1$ and $x = m/2^n \in [0,1]$. For every real h with $|h| \le 2^{-N-1}$,*

$$\boxed{G_N(x+h) = \sum_{k=0}^{\lfloor n/2 \rfloor} e_k\, 2^{-2k(N+1)} \sum_{r=0}^{n-2k} \frac{F_{\mathrm{bd}}^{(r+2k)}(x)}{r!} h^r\,,} \tag{86}$$

with the finite reciprocal coefficients e_k of Equation (53). Equivalently, on this cell $G_N(x+h) = [M(2^{-N-1}\mathrm{D}_h)]^{-1} P_x(h)$, the inverse meaning its finite action on polynomials of degree at most n.

Proof. Write $R = N-n$. The centred prefix splits into independent parts as $Y_N = X_n + 2^{-n} Y_R$ with $Y_R = (1+Z_R)/2$, $Y_0 = \frac{1}{2}$, and Y_R is supported in $[2^{-R-1}, 1-2^{-R-1}]$. The bound on h gives $|2^n h| \le 2^{-R-1}$, so in the box formula for the first n variables the cutoff pattern cannot change: all indices $\ell < m$ contribute full powers and all $\ell \geq m$ contribute zero (at the two cell endpoints some arguments can equal zero, which changes nothing since $n \geq 1$). Thus

$$G_N(x+h) = \frac{2^{-T_n}}{n!} \sum_{\ell=0}^{m-1} \varepsilon_\ell \, \mathbb{E}\big(2m - 2\ell - 1 + 2^{n+1}h - Z_R\big)^n. \tag{87}$$

Replacing Z_R by Z produces a polynomial in h whose rth derivative at 0 is Equation (82); that polynomial is exactly $P_x(h)$. For any polynomial $p(v)$, symmetry gives $\mathbb{E}p(v - Z_R) = M_R(\mathrm{D}_v)p(v)$ and $\mathbb{E}p(v - Z) = M(\mathrm{D}_v)p(v)$ (Theorem 7.2 with $\eta = 1$, or directly by Taylor expansion), and $M_R(t) = M(t)/M(2^{-R}t)$. These identities are legitimate in the finite-dimensional space of polynomials. In Equation (87) the variable is $v = 2m - 2\ell - 1 + 2^{n+1}h$, so $\mathrm{D}_v = 2^{-n-1}\mathrm{D}_h$ and the denominator is $M(2^{-R-n-1}\mathrm{D}_h) = M(2^{-N-1}\mathrm{D}_h)$. Expanding its reciprocal by Equation (52) gives Equation (86). □

Corollary 7.4 (Polynomial dependence on the squared tail scale)**.** *For a fixed dyadic $x = m/2^n \in [0,1]$, $n \geq 1$, $d = \lfloor n/2 \rfloor$, and all integers $N \geq n$,*

$$G_N(x) = Q_x(4^{-N}), \qquad Q_x(z) := \sum_{k=0}^{d} e_k \, 4^{-k} \, F_{\mathrm{bd}}^{(2k)}(x) \, z^k, \tag{88}$$

so $Q_x(0) = F_{\mathrm{bd}}(x)$ and $\deg Q_x \leq d$.

This is an exact identity for the whole tail of the sequence $(G_N(x))_{N \geq n}$, not a finite-order asymptotic expansion; its constant term can therefore be extracted from finitely many depths. At $x = \frac{1}{4}$, where $F'_{\mathrm{bd}} = 1$, $F''_{\mathrm{bd}} = 8$ and the higher derivatives vanish,

$$G_N\big(\tfrac{1}{4} + h\big) = \frac{5}{72} + h + 4h^2 - \frac{1}{9} 4^{-N} \qquad (N \geq 2, \ |h| \leq 2^{-N-1}), \tag{89}$$

which, after the index shift of Section 8.5, is the quarter-point cell that is kernel-verified in the repository (Section D).

Remark 7.5 (The degree drop seen through Prouhet cancellation). That the polynomial piece of G_N on the cell has degree at most n, although a general piece has degree N, can also be seen combinatorially. The finite product Equation (63) has a zero of order L at $z = 1$, so differentiating it at $z = 1$ gives Prouhet's cancellation

$$\sum_{r=0}^{2^L - 1} \varepsilon_r \, r^i = 0 \qquad (0 \leq i < L). \tag{90}$$

With $N = n + L$ and the active indices $h = b2^L + r$, $0 \leq b < m$, $0 \leq r < 2^L$, the signs factor as $\varepsilon_h = \varepsilon_b \varepsilon_r$, and each block of the cell polynomial is a signed sum of powers $(A + 2^{n+1}h' - Br)^{n+L}$ in the cell variable h'; the coefficient of h'^i is a polynomial in r of degree $n + L - i$, which is annihilated by Equation (90) when $i > n$. This gives the degree bound without the tail; the deconvolution then identifies the coefficients.

7.5 Why the degree bound is attained

Lemma 7.6 (Signs of the reciprocal coefficients)**.** *For every $k \geq 0$, $(-1)^k e_k > 0$.*

Proof. Euler's product for the sine [9, Eq. 4.22.1] gives, near $t = 0$,

$$\frac{1}{M(\mathrm{i}t)} = \prod_{j=1}^{\infty} \prod_{\ell=1}^{\infty} \Big(1 - \frac{t^2}{4^j \pi^2 \ell^2}\Big)^{-1}. \tag{91}$$

The positive numbers $(4^j \pi^2 \ell^2)^{-1}$ have finite sum, so the product converges near zero and its power-series coefficients are limits of those of finite products. Every coefficient of a finite product of geometric series $\sum_i(\alpha t^2)^i$ with $\alpha > 0$ is nonnegative, and for each degree k even one factor contributes a strictly positive coefficient; hence the coefficient of t^{2k} on the left, which is $(-1)^k e_k$, is strictly positive. The infinite product is used only for the sign; the evaluation formula for e_k remains the finite Equation (53). ☐

Proposition 7.7 (Exact degree at a reduced interior dyadic). *If $n \geq 1$, $0 < m < 2^n$ and m is odd, then* $\deg Q_{m/2^n} = \lfloor n/2 \rfloor$.

Proof. By Equation (22) and Equation (23): if $n = 2d$, the top even derivative is $F_{\mathrm{bd}}^{(2d)}(m/2^{2d}) = 2^{T_{2d}} F_{\mathrm{gl}}(m) = 2^{T_{2d}} \varepsilon_{(m-1)/2} \neq 0$; if $n = 2d+1$, it is $F_{\mathrm{bd}}^{(2d)}(m/2^{2d+1}) = 2^{T_{2d}} F_{\mathrm{gl}}(m/2) = 2^{T_{2d}-1} \varepsilon_{\lfloor m/4 \rfloor} \neq 0$, which also covers $n = 1$, $d = 0$. The coefficient of z^d in Equation (88) is then nonzero by Theorem 7.6. ☐

A dyadic argument should therefore be reduced before choosing the number of levels: an unreduced depth gives a correct formula with a correct upper bound, but introduces unnecessary interpolation nodes.

8 Quarter-base extraction and its sharpness

8.1 Extraction from arbitrary distinct depths

Theorem 8.1 (Extraction from arbitrary depths). *Fix $n \geq 1$, $d = \lfloor n/2 \rfloor$, and distinct integers $N_0, \ldots, N_d \geq n$. For every $x = m/2^n \in [0,1]$,*

$$F_{\mathrm{bd}}(x) = \sum_{i=0}^{d} W_i \, G_{N_i}(x), \qquad W_i = \prod_{\substack{0 \leq \ell \leq d \\ \ell \neq i}} \frac{4^{-N_\ell}}{4^{-N_\ell} - 4^{-N_i}}. \tag{92}$$

The weights depend only on the selected depths, not on m.

Proof. Let $z_i = 4^{-N_i}$. For any polynomial Q of degree at most d, $Q(z) = \sum_i Q(z_i) \prod_{\ell \neq i}(z - z_\ell)/(z_i - z_\ell)$: the difference of the two sides has degree at most d and vanishes at the $d+1$ distinct points z_i, so it is zero. Evaluate at $z = 0$ with $Q = Q_x$ from Theorem 7.4. ☐

8.2 Consecutive depths and the Gaussian-binomial row

For $0 < q < 1$ and $0 \leq j \leq d$ define

$$w_{d,j}(q) := \prod_{\substack{0 \leq \ell \leq d \\ \ell \neq j}} \frac{-q^\ell}{q^j - q^\ell}, \tag{93}$$

and $w_{d,j} := w_{d,j}(\frac{1}{4})$.

Lemma 8.2 (Constant extraction on a geometric grid). *For every polynomial P of degree at most d and every $z \neq 0$,*

$$P(0) = \sum_{j=0}^{d} w_{d,j}(q) \, P(zq^j), \tag{94}$$

equivalently $\sum_{j=0}^{d} w_{d,j}(q) q^{jr} = [r = 0]$ for $0 \leq r \leq d$. In closed form,

$$\boxed{w_{d,j}(q) = \frac{(-1)^{d-j} q^{T_{d-j}}}{(q;q)_j \, (q;q)_{d-j}} = \frac{(-1)^{d-j} q^{T_{d-j}}}{(q;q)_d} \begin{bmatrix} d \\ j \end{bmatrix}_q,} \tag{95}$$

and the generating polynomial of the row is

$$\sum_{j=0}^{d} w_{d,j}(q)\, v^j = \prod_{r=1}^{d} \frac{v-q^r}{1-q^r}. \tag{96}$$

At $q = \frac{1}{4}$, $(q;q)_r = 4^{-T_r} \prod_{s=1}^{r}(4^s - 1)$, *which turns* Equation (95) *into the integer-product form* Equation (5).

Proof. The Lagrange basis polynomial for the node zq^j among the nodes $zq^0, \ldots, zq^d$, evaluated at 0, is $\prod_{\ell \neq j}(0 - zq^\ell)/(zq^j - zq^\ell)$, in which z cancels; that is Equation (93), and Lagrange interpolation (the uniqueness argument of the previous proof) gives Equation (94); the monomials $P = z^r$ give the moment identities. Splitting the product into $\ell < j$, where the factor is $1/(1 - q^{j-\ell})$, and $\ell > j$, where it is $-q^{\ell-j}/(1 - q^{\ell-j})$, gives Equation (95), the exponent being $\sum_{i=1}^{d-j} i = T_{d-j}$. The generating polynomial has degree d, vanishes at $v = q^r$ for $1 \le r \le d$ by the moment identities, and takes the value 1 at $v = 1$ (the case $r = 0$), which determines it as the right side of Equation (96). The closed forms are also consequences of the finite q-binomial theorem [9, §17.2], but no such theorem is needed. □

Theorem 8.3 (Extraction at consecutive depths). *For* $n \ge 1$, $0 \le m \le 2^n$, $d = \lfloor n/2 \rfloor$ *and every integer* $R \ge 0$,

$$\boxed{F_{\mathrm{bd}}\left(\frac{m}{2^n}\right) = \sum_{j=0}^{d} w_{d,j}\, G_{n+R+j}\left(\frac{m}{2^n}\right),} \tag{97}$$

and substituting Equation (81) *gives* Equation (7), *hence* Equation (6) *at* $R = 0$. *In fully explicit* q*-notation, with* $Q = \frac{1}{4}$,

$$\boxed{\begin{aligned} F_{\mathrm{bd}}\left(\frac{m}{2^n}\right) &= \frac{1}{(Q;Q)_d} \sum_{j=0}^{d} \frac{(-1)^{d-j} Q^{T_{d-j}}}{2^{T_{n+j}}\,(n+j)!} \begin{bmatrix} d \\ j \end{bmatrix}_Q \\ &\quad \times \sum_{h=0}^{2^j m - 1} \varepsilon_h \left(2^{j+1}m - 2h - 1\right)^{n+j}. \end{aligned}} \tag{98}$$

No moments, Bernoulli numbers, derivative values, recursively defined coefficients, limits, or coefficient extractors occur in this formula.

Proof. Apply Theorem 8.2 to the polynomial Q_x of Theorem 7.4 with $q = \frac{1}{4}$ and $z = 4^{-(n+R)}$, whose nodes are the depths $n + R + j$; equivalently, this is Theorem 8.1 with $N_j = n + R + j$, the common factor $4^{-(n+R)}$ cancelling from the weights. Then replace each $G_{n+R+j}(m/2^n)$ by the integer sum Equation (81) with $A = 2^{R+j}m$, and at $R = 0$ insert Equation (95). The case $d = 0$ uses the single weight one. □

The row Equation (6) uses $d+1$ splines of orders $n, \ldots, n+d$; its largest power is $n+d$, and before simplification it contains $\sum_{j=0}^{d} 2^j m = m(2^{d+1} - 1)$ signed integer powers. It is attractive when the absence of every auxiliary coefficient sequence is more important than minimizing arithmetic; for large numerators the block formula is usually the better organization. Consecutive depths need not start at n (the parameter R), and arbitrary distinct depths $\ge n$ are covered by Theorem 8.1, which permits the reuse of already available spline values.

8.3 A precise optimality statement

The number $d + 1$ is not a lower bound on all possible algorithms — the binary formulas are not interpolation rules — but it is sharp within the natural class of grid-wide linear rules.

Theorem 8.4 (Minimal number of samples for a grid-wide linear rule). *Fix $n \geq 1$ and $d = \lfloor n/2 \rfloor$. Suppose distinct depths $N_1, \ldots, N_L \geq n$ and real numbers $c_1, \ldots, c_L$, independent of m, satisfy $F_{\mathrm{bd}}(m/2^n) = \sum_{i=1}^{L} c_i G_{N_i}(m/2^n)$ for every $m = 0, \ldots, 2^n$. Then $L \geq d+1$.*

Proof. The grid contains $x_0 = 1$ and $x_r = 2^{-2r}$ for $1 \leq r \leq d$. The polynomial Q_{x_0} is the constant 1, and by Theorem 7.7 Q_{x_r} has degree exactly r. These $d+1$ polynomials therefore form a basis of the polynomials of degree at most d: their leading terms make the coefficient matrix triangular with nonzero diagonal. The supposed rule recovers $Q(0)$ from the values $Q(z_i)$, $z_i = 4^{-N_i} > 0$, for each basis element, hence by linearity for every polynomial Q of degree at most d. If $L \leq d$, take $Q(z) = \prod_{i=1}^{L}(z - z_i)$: all sampled values vanish but $Q(0) \neq 0$, a contradiction; $L = 0$ is included by the empty product. □

The theorem does not exclude weights depending on the point, nonlinear formulas, other kinds of samples, or a direct finite coefficient calculation.

8.4 The extraction weights are well conditioned

Proposition 8.5 (Absolute row sum). *For $0 < q < 1$,*

$$\sum_{j=0}^{d} |w_{d,j}(q)| = \frac{(-q;q)_d}{(q;q)_d} = \prod_{r=1}^{d} \frac{1+q^r}{1-q^r} \leq \prod_{r=1}^{\infty} \frac{1+q^r}{1-q^r} < \infty. \tag{99}$$

For $q = \frac{1}{4}$ the absolute sum is strictly less than 2, uniformly in d; the limiting values are $1.96926\ldots$ at $q = \frac{1}{4}$ and $8.25599\ldots$ at $q = \frac{1}{2}$. No nonnegative extraction row exists when $d \geq 1$.

Proof. The sign of $w_{d,j}(q)$ is $(-1)^{d-j}$, so the absolute sum is $(-1)^d \sum_j w_{d,j}(q)(-1)^j$, which by Equation (96) at $v = -1$ equals $\prod_r (1+q^r)/(1-q^r)$; this is the first equality, and $\prod_{r \leq d}(1+q^r) = (-q;q)_d$. The infinite product is finite because $\log((1+u)/(1-u)) \leq 2u/(1-u)$ for $0 \leq u < 1$ and $\sum_r q^r$ converges. For $q = \frac{1}{4}$ the first factor is $\frac{5}{3}$ and $\sum_{r \geq 2} \log \frac{1+4^{-r}}{1-4^{-r}} \leq \frac{32}{15} \sum_{r \geq 2} 4^{-r} = \frac{8}{45}$, while $\log \frac{6}{5} \geq \frac{1}{5} - \frac{1}{50} = \frac{9}{50} > \frac{8}{45}$; hence the product is below $\frac{5}{3} \cdot \frac{6}{5} = 2$ for every d (for $d = 0$ the sum is 1). Finally, positive nodes with nonnegative weights summing to one cannot have a vanishing first moment, so a nonnegative row is impossible when $d \geq 1$. □

If each exactly computed input $G_{N_j}(x)$ carries an absolute error at most δ, combining them with the exact weights adds error below 2δ. This bounds the amplification by the outer filter only; the alternating power sum inside an individual spline can cancel severely, and exact integer or rational arithmetic is the appropriate default there.

8.5 Dictionary to the canonical extraction volume

The canonical volume of this subgroup [2] works with the density splines up_n, the inverse Fourier transforms of the prefix products with factors $k = 0, \ldots, n$ of $\prod_{k \geq 0} \mathrm{sinc}_{\mathrm{rad}}(\pi\xi/2^k)$, and with the reciprocal-product coefficients a_r^{vol} defined by $1/\widehat{\mathrm{up}}_{2\pi}(\xi) = \sum_r a_r^{\mathrm{vol}}(2\pi\xi)^{2r}$. In the notation of this document,

$$\mathrm{up}_n = u_{n+1}, \qquad a_r^{\mathrm{vol}} = (-1)^r e_r, \qquad G_N(x) = \mathrm{up}_{N-1}\text{'s distribution function} = F_{\mathrm{bd}}\text{-side of } u_N, \tag{100}$$

where u_N is the density of Z_N (Section 11.2); indeed $\widehat{\mathrm{up}}_{2\pi}(\xi) = M(2\pi\mathrm{i}\xi)$, so $1/\widehat{\mathrm{up}}_{2\pi}(\xi) = \sum_k e_k(2\pi\mathrm{i}\xi)^{2k} = \sum_k (-1)^k e_k(2\pi\xi)^{2k}$, and the volume's $a_1^{\mathrm{vol}} = \frac{1}{18}$, $a_2^{\mathrm{vol}} = \frac{31}{16200}$, $a_3^{\mathrm{vol}} = \frac{2347}{42865200}$ are the absolute values of Equation (54). The volume's main theorem, $\mathrm{up}_n(x) = \mathrm{up}(x) + \sum_{r=1}^{d}(-1)^r a_r^{\mathrm{vol}} 4^{-r}\,\mathrm{up}^{(2r)}(x) 4^{-rn}$ for $n \geq s(x)$, is Equation (116) below read through this dictionary: $u_N(x) = \sum_k e_k 4^{-Nk}\,\mathrm{up}^{(2k)}(x)$ with $N = n+1$. Its extraction row, with the weights $(-1)^{d-j} Q^{\binom{d-j+1}{2}} \left[{d \atop j}\right]_Q / (Q;Q)_d$, is the row Equation (95) of the present document, and its depth $s(x)$

of a dyadic $x \in [-1,1]$ is the reduced denominator exponent used here. The volume also proves, from Rvachev's functional-differential equation, that the defect at the last level before the onset is $-\varepsilon_k 2^{-\binom{s}{2}} B_s/s!$, and records which pieces are kernel-verified; Section D collects the crosswalks relevant to the formulas of this document.

9 The half-base formula and the general shift

The quarter-base formula is not obtained by replacing $\frac{1}{2}$ by $\frac{1}{4}$ in an existing identity: centring removes all odd cutoff powers and halves the interpolation degree. A shift parameter makes this precise. For $0 \le \alpha \le 1$ set

$$G_{N,\alpha}(x) := \mathbb{P}(X_N + \alpha 2^{-N} \le x), \tag{101}$$

so $G_{N,1/2} = G_N$ and $G_{N,0} = H_N$. For $x = m/2^n$, $N \ge n \ge 1$ and $A = 2^{N-n}m$, the box formula gives

$$G_{N,\alpha}\Big(\frac{m}{2^n}\Big) = \frac{2^{-\binom{N}{2}}}{N!} \sum_{h=0}^{A-1} \varepsilon_h \, (A-h-\alpha)^N; \tag{102}$$

at $\alpha \in \{0,1\}$ the extra boundary terms have vanishing Nth power, so the same expression covers both endpoints. For arbitrary complex α define $G_{N,\alpha}(m/2^n)$ by the polynomial on the right; it loses its probability interpretation but remains a finite algebraic object.

Proposition 9.1 (General shift and the half-base row). *For $n \ge 1$, $0 \le m \le 2^n$ and every $\alpha \in \mathbb{C}$,*

$$F_{\mathrm{bd}}\Big(\frac{m}{2^n}\Big) = \sum_{j=0}^{n} w_{n,j}\big(\tfrac{1}{2}\big)\, G_{n+j,\alpha}\Big(\frac{m}{2^n}\Big), \tag{103}$$

equivalently

$$\boxed{\begin{aligned} F_{\mathrm{bd}}\Big(\frac{m}{2^n}\Big) &= \frac{1}{2^{n^2}\,\big(\frac{1}{2};\frac{1}{2}\big)_n} \sum_{j=0}^{n} \frac{\binom{n}{j}_{1/2}}{2^{j(j-1)}\,(n+j)!} \\ &\quad \times \sum_{h=0}^{2^j m-1} \varepsilon_h \, \big(h - 2^j m + \alpha\big)^{n+j}. \end{aligned}} \tag{104}$$

Proof. First let $0 \le \alpha \le 1$ and $\eta = 2^{-N}$. The head–tail identity reads $F_{\mathrm{bd}}(t) = \mathbb{E}G_{N,\alpha}(t - \eta(X' - \alpha))$, and at $x = A\eta$ the whole averaging interval $[x - \eta(1-\alpha), x + \eta\alpha]$ lies in one polynomial cell of $G_{N,\alpha}$, whose endpoints are consecutive possible knots. Apply Theorem 7.2 with $W = X' - \alpha$ (supported in $[-1,1]$, not symmetric) and the degree N: since the jet of F_{bd} at x terminates at order n,

$$G_{N,\alpha}(x) = \sum_{r=0}^{n} \gamma_r(\alpha)\, \eta^r F_{\mathrm{bd}}^{(r)}(x), \qquad \gamma_r(\alpha) = [z^r]\, \frac{1}{M_W(-z)},$$

with $\gamma_0 = 1$ and coefficients independent of N. Thus $G_{N,\alpha}(x)$ is an exact polynomial of degree at most n in 2^{-N} with constant term $F_{\mathrm{bd}}(x)$, and Theorem 8.2 at $q = \frac{1}{2}$ with $n+1$ consecutive depths gives Equation (103). When Equation (102) is substituted, both sides are polynomials in α; equality on $[0,1]$ implies equality for all complex α. Finally insert the weights Equation (95) at $q = \frac{1}{2}$ and write $(A-h-\alpha)^{n+j} = (-1)^{n+j}(h-A+\alpha)^{n+j}$, which cancels the sign $(-1)^{n-j}$ of the weight; the powers of two combine as $T_{n-j} + \binom{n+j}{2} = n^2 + j(j-1)$, giving Equation (104). ☐

Formula Equation (104) is the unit-interval form of the expression posed by Reshetnikov in 2019 [7] and subsequently developed in the repository [1], which also treats the signed extension at unrestricted numerators; the fold Equation (119) evaluates it everywhere without reproving that identity. At $\alpha = \frac{1}{2}$

the variable $W = Z'/2$ is symmetric, the odd coefficients γ_r vanish, and the degree drops from at most n in 2^{-N} to at most $\lfloor n/2 \rfloor$ in 4^{-N}: this is precisely the passage from Equation (103) to Equation (97), the half-base row using $n+1$ splines of orders up to $2n$ where the quarter-base row uses $d+1$ of orders up to $n+d$.

10 An integer determinant, a common denominator, and certified rounding

10.1 The bordered determinant

The composition and partition formulas expose individual coefficients. A bordered integer matrix packages the entire dyadic value into a single ratio of integers; its Leibniz expansion is a finite sum of products, so no matrix inversion or recurrence is part of the definition.

Put $d = \lfloor n/2 \rfloor$ and define the integer row entries

$$t_j := \binom{n}{2j} \mathcal{S}_{n-2j}(m) = \binom{n}{2j} \sum_{h=0}^{m-1} \varepsilon_h \, (2m-2h-1)^{n-2j} \qquad (0 \le j \le d), \tag{105}$$

with $\mathcal{S}_0(m) = \sum_{h<m} \varepsilon_h$. Let L_d be the $d \times d$ lower triangular integer matrix with indices $1 \le k, j \le d$ and entries

$$(L_d)_{kj} = \begin{cases} (2k+1)(4^k-1), & j = k, \\ -\binom{2k+1}{2j}, & j < k, \\ 0, & j > k, \end{cases} \tag{106}$$

and form the bordered $(d+1) \times (d+1)$ matrix

$$B_{n,m} := \begin{pmatrix} & L_d & & \mathbf{1} \\ -t_1 & \cdots & -t_d & t_0 \end{pmatrix}, \tag{107}$$

where $\mathbf{1}$ is the column of d ones; for $d = 0$ this is the 1×1 matrix (t_0). Every entry is an explicit integer.

Theorem 10.1 (Integer determinant formula). *For $n \ge 0$ and $0 \le m \le 2^n$,*

$$\boxed{F_{\mathrm{bd}}\Big(\frac{m}{2^n}\Big) = \frac{\det B_{n,m}}{D_n}, \qquad D_n := 2^{T_n}\, n! \prod_{k=1}^{d} (2k+1)(4^k-1) = 2^{T_n}\, n!\, (2d+1)!! \prod_{k=1}^{d} (4^k-1).} \tag{108}$$

In particular the reduced denominator of every value on the grid of spacing 2^{-n}, and of every Rvachev value $\mathrm{up}(m/2^n)$, divides D_n.

Proof. Multiplying Equation (51) by $(2k+1)!$ and separating the top term gives, in terms of the moments $c_j = (2j)!a_j$,

$$(2k+1)(4^k-1)\, c_k = \sum_{j=0}^{k-1} \binom{2k+1}{2j} c_j \qquad (k \ge 1),$$

and since $c_0 = 1$ this is the finite system $L_d(c_1, \ldots, c_d)^\top = \mathbf{1}$. The diagonal entries are nonzero and $\det L_d = \prod_{k=1}^{d} (2k+1)(4^k-1)$. For a block matrix with invertible upper-left block, $\det \begin{pmatrix} L & \mathbf{1} \\ -t^\top & t_0 \end{pmatrix} = \det L \cdot \big(t_0 + t^\top L^{-1} \mathbf{1}\big)$ (the Schur complement; equivalently, add $t^\top L^{-1}$ times the first d rows to the last row, which clears its first d entries). With $L_d^{-1}\mathbf{1} = c = (c_1, \ldots, c_d)^\top$ this gives

$$\det B_{n,m} = \det L_d \Big(t_0 + \sum_{j=1}^{d} t_j c_j\Big).$$

By Equation (35), $F_{\mathrm{bd}}(m/2^n) = 2^{-T_n}(n!)^{-1}\sum_{j=0}^{d} t_j c_j$, which is Equation (108); the case $d = 0$ is the same formula directly. The moment relation was used only to identify the matrix; the final expression asks for no moment. Integrality of the entries gives the denominator statement for F_{bd}, and the fold Equation (15) for up. □

In the literal finite-sum reading,

$$\det B_{n,m} = \sum_{\sigma\in\mathfrak{S}_{d+1}} (-1)^{\#\{(i,j):\, i<j,\ \sigma(i)>\sigma(j)\}} \prod_{i=1}^{d+1} (B_{n,m})_{i,\sigma(i)}, \tag{109}$$

so even the determinant notation can be removed; many terms vanish because L_d is triangular, and an implementation should use exact elimination rather than this factorial-size expansion. For $n = 2$, $m = 1$: $B_{2,1} = \begin{pmatrix} 9 & 1 \\ -1 & 1 \end{pmatrix}$, $\det B_{2,1} = 10$, $D_2 = 2^3 \cdot 2! \cdot 9 = 144$, and $F_{\mathrm{bd}}(\frac{1}{4}) = \frac{10}{144} = \frac{5}{72}$.

Remark 10.2 (An equivalent $(d+2)$-bordering). One may also border the full $(d+1)\times(d+1)$ triangular matrix T with $T_{00} = 1$, $T_{ii} = (2i+1)(4^i-1)$, $T_{ij} = -\binom{2i+1}{2j}$ for $j < i$, by the unit vector e_0 and the row $v = (t_0, \dots, t_d)$: $F_{\mathrm{bd}}(m/2^n) = -\det\begin{pmatrix} T & e_0 \\ v^\mathsf{T} & 0 \end{pmatrix} / D_n$, by the same Schur-complement identity $\det = -\det T \cdot v^\mathsf{T} T^{-1} e_0 = -\det T \cdot v^\mathsf{T} c$. The $(d+1)$-form above absorbs the row $c_0 = 1$ into the border.

10.2 A second denominator bound, and the prefix error

D_n is a convenient common denominator, not a claim of minimality, and it does not replace the sharper arithmetic of [5]. A different elementary bound follows directly from the positive composition formula.

Proposition 10.3 (Denominator bound from compositions). *For $n \geq 0$ and $d = \lfloor n/2 \rfloor$, the number $D'_n := 2^{T_n}(n+d)!\prod_{j=1}^{d}(4^j-1)$ satisfies $D'_n F_{\mathrm{bd}}(m/2^n) \in \mathbb{Z}$ for $0 \leq m \leq 2^n$.*

Proof. Use Equation (50). A summand with index $k \leq d$ and composition $k = r_1 + \cdots + r_\ell$ has denominator dividing $2^{T_n}(n-2k)!\prod_i (2r_i+1)!\prod_i (4^{R_i}-1)$. The suffix sums R_i are distinct integers in $[1,k]$, so the last product divides $\prod_{j\leq d}(4^j-1)$. The factorial arguments add up to $(n-2k) + \sum_i (2r_i+1) = n + \ell \leq n + d$, and by integrality of multinomial coefficients their product divides $(n+\ell)!$, which divides $(n+d)!$; the term $k = 0$ has just $n!$. Every summand, hence their integer-signed sum, has denominator dividing D'_n. □

Neither bound dominates the other in general: $D_4 = 16588800$ while $D'_4 = 2^{10}\cdot 720 \cdot 45 = 33177600$, but D'_n has the smaller factorial for large n relative to the double factorial. Either may be used below.

Lemma 10.4 (Uniform prefix error). *For every $N \geq 1$ and every real x, $|G_N(x) - F_{\mathrm{bd}}(x)| \leq 2^{-N}$.*

Proof. Couple the variables by $X = Y_N + 2^{-N-1}Z'$ with Z' an independent copy of Z and $|Z'| \leq 1$. With $\delta = 2^{-N-1}$ this gives $F_{\mathrm{bd}}(x-\delta) \leq G_N(x) \leq F_{\mathrm{bd}}(x+\delta)$, and the 2-Lipschitz bound of Theorem 2.1 proves the claim. □

Corollary 10.5 (Certified one-prefix rounding). *For $n \geq 1$ let D be any of the common denominators D_n, D'_n, and choose the explicit integer $N = \max\{n,\ 2 + \lfloor \log_2 D \rfloor\}$, so that $2^N > 2D$. Then*

$$\boxed{F_{\mathrm{bd}}\left(\frac{m}{2^n}\right) = \frac{1}{D}\left\lfloor \frac{D\,\mathcal{S}_N(2^{N-n}m)}{2^{T_N} N!} + \frac{1}{2} \right\rfloor.} \tag{110}$$

Proof. By Theorem 10.4, $|D\,G_N(x) - D\,F_{\mathrm{bd}}(x)| \leq D2^{-N} < \frac{1}{2}$, and $DF_{\mathrm{bd}}(x)$ is an integer; nearest-integer rounding is therefore exact, with no tie. Substitute Equation (81). □

The logarithm in the choice of N can be replaced by the binary digit length of the explicitly specified integer D. The formula has only one finite prefix, but that prefix is enormous; it is a mathematical certification device, not a recommended implementation.

11 Rvachev values, the signed extension, derivatives and primitives

11.1 Every dyadic bump value

For an integer a and $n \geq 0$, the support and the fold Equation (15) give

$$\boxed{\operatorname{up}\left(\frac{a}{2^n}\right) = \begin{cases} F_{\text{bd}}\left(\dfrac{2^n - |a|}{2^n}\right), & |a| \leq 2^n, \\ 0, & |a| > 2^n, \end{cases}} \tag{111}$$

so every formula for F_{bd} gives a closed rational formula for up with no new coefficients; in particular $\operatorname{up}(0) = 1$, $\operatorname{up}(\pm 1) = 0$, $\operatorname{up}(\pm\frac{1}{2}) = \frac{1}{2}$, and $\operatorname{up}(\frac{3}{4}) = F_{\text{bd}}(\frac{1}{4}) = \frac{5}{72}$.

11.2 Direct extraction from finite densities

When the available data are finite sinc products rather than distribution functions, a direct formula is useful. Let u_N be the density of $Z_N = \sum_{j=1}^{N} 2^{-j} V_j$; under the 2π convention

$$\widehat{u_N}_{2\pi}(\xi) = \prod_{k=0}^{N-1} \operatorname{sinc}_{\text{rad}}\left(\frac{\pi\xi}{2^k}\right), \qquad \operatorname{sinc}_{\text{rad}}(z) = \frac{\sin z}{z}, \tag{112}$$

so a product ending with the factor $k = n$ is u_{n+1} in this notation; this one-factor offset matters in exact cutoff formulas (Section 8.5). Since $Z_N = \sum_{j=1}^{N} 2^{1-j} U_j - (1 - 2^{-N})$, differentiating the weighted-uniform distribution function of Theorem 3.1 once gives, for $N \geq 2$ (so that the density is continuous),

$$u_N(x) = \frac{2^{N(N-1)/2}}{(N-1)!} \sum_{h=0}^{2^N-1} \varepsilon_h \left(x + 1 - 2^{-N} - 2^{1-N} h\right)_+^{N-1}, \tag{113}$$

whose knots are the odd multiples of 2^{-N}; for $N = 1$ the same display with the convention $s_+^0 = [s > 0]$ is the indicator density of $[-\frac{1}{2}, \frac{1}{2}]$. For $|a| < 2^n$ and $N > n \geq 0$ put $A = 2^N + a2^{N-n}$, a positive even integer; the terms with $2h < A - 1$ are exactly the active ones, and

$$\boxed{u_N\left(\frac{a}{2^n}\right) = \frac{1}{(N-1)!\, 2^{N(N-1)/2}} \sum_{h=0}^{A/2-1} \varepsilon_h \left(A - 1 - 2h\right)^{N-1}.} \tag{114}$$

The two finite objects are related by an exact fold: the independent-copy identity $Z_N \overset{\text{law}}{=} (V + Z'_{N-1})/2$ gives $u_N(x) = \mathbb{P}(2x - 1 \leq Z'_{N-1} \leq 2x + 1)$, and for $x \geq 0$ the upper cutoff may be discarded, so by symmetry

$$u_N(x) = G_{N-1}\big(1 - |x|\big) \qquad (x \in \mathbb{R},\ N \geq 2). \tag{115}$$

Theorem 11.1 (Direct quarter-base extraction for the bump). *Let $n \geq 0$, $|a| < 2^n$, $d = \lfloor n/2 \rfloor$, and $x = a/2^n$. Then for every $N > n$*

$$u_N(x) = \sum_{k=0}^{d} e_k\, 4^{-Nk} \operatorname{up}^{(2k)}(x), \tag{116}$$

and for every $R \geq 0$

$$\boxed{\operatorname{up}\left(\frac{a}{2^n}\right) = \sum_{j=0}^{d} w_{d,j}\, u_{n+1+R+j}\left(\frac{a}{2^n}\right),} \tag{117}$$

with each u_N the finite integer sum Equation (114). For $|a| \geq 2^n$ the bump value is zero.

Proof. With $\eta = 2^{-N}$, $Z \overset{\text{law}}{=} Z_N + \eta Z'$ gives $\operatorname{up}(t) = \mathbb{E} u_N(t - \eta Z')$. Since $N > n$, the number $x/\eta = a2^{N-n}$ is even while the knots of u_N are odd multiples of η, so $[x - \eta, x + \eta]$ is one polynomial cell, on which u_N has degree $N - 1$. Apply Theorem 7.2 with this degree, $W = Z'$ and η: the jet of up at x terminates at order n (Theorem 2.2), only even orders survive by symmetry, and the reciprocal operator $[M(-\eta \mathrm{D})]^{-1} = \sum_k e_k \eta^{2k} \mathrm{D}^{2k}$ evaluated at $s = 0$ gives Equation (116). Constant extraction on the geometric grid 4^{-N}, $N = n + 1 + R + j$, with $q = \frac{1}{4}$ yields Equation (117). The case $N = 1$, $n = 0$, $a = 0$ is the constant density on its interior and follows by the same degree-zero argument; outside $[-1, 1]$ and at its endpoints the bump vanishes. Alternatively, apply Theorem 8.3 at $(2^n - |a|)/2^n$ with depths $n + R + j$ and fold back by Equation (115), which explains the shift by one in the levels. □

At $x = \frac{1}{4}$: $u_3(\frac{1}{4}) = \frac{15}{16}$, $u_4(\frac{1}{4}) = \frac{179}{192}$, and $\operatorname{up}(\frac{1}{4}) = \big(4u_4 - u_3\big)/3 = \frac{67}{72}$; indeed $\operatorname{up}''(\frac{1}{4}) = -8$, so the whole tail satisfies $u_N(\frac{1}{4}) = \frac{67}{72} + \frac{4}{9}4^{-N}$ for $N \geq 3$. Two suitable consecutive values determine the limit with zero remainder, unlike numerical extrapolation from an asymptotic expansion.

11.3 The signed global extension at every dyadic argument

For $a \geq 0$ define

$$b := \left\lfloor \frac{a}{2^{n+1}} \right\rfloor, \qquad r := a - 2^{n+1} b, \qquad a^* := \min\left\{r,\, 2^{n+1} - r\right\}. \tag{118}$$

Then $0 \leq a^* \leq 2^n$ and, by Equation (21) and the fold of the bump around its centre,

$$\boxed{F_{\mathrm{gl}}\left(\frac{a}{2^n}\right) = \varepsilon_b\, F_{\mathrm{bd}}\left(\frac{a^*}{2^n}\right),} \qquad F_{\mathrm{gl}}(x) = 0 \;\; (x < 0). \tag{119}$$

For instance $F_{\mathrm{bd}}(3) = 1$ whereas $F_{\mathrm{gl}}(3) = -1$. The block formula Equation (61) performs this reduction implicitly through the high-bit signs η_j and needs no separate fold. No formula requires $a/2^n$ to be reduced: directly from the proved identities any of the closed expressions satisfies $\mathcal{F}_n(a) = \mathcal{F}_{n+1}(2a)$, and reducing merely shortens the sums; the reflection $F_{\mathrm{bd}}(m/2^n) + F_{\mathrm{bd}}((2^n - m)/2^n) = 1$ is a further useful reduction.

11.4 Closed forms for every dyadic derivative

For $a, n \geq 0$, Equation (22) and Equation (23) give

$$F_{\mathrm{gl}}^{(r)}\left(\frac{a}{2^n}\right) = \begin{cases} 2^{T_r}\, F_{\mathrm{gl}}(a/2^{n-r}), & 0 \leq r \leq n, \\ 0, & r \geq n+1, \end{cases} \tag{120}$$

since for $r \geq n + 1$ the argument $2^r a/2^n$ is an even integer. On the interior of $[0, 1]$ the derivatives of F_{bd} agree with those of F_{gl}, and at 0 and 1 all positive-order derivatives of F_{bd} vanish. For the bump use the signed extension without an absolute-value fold:

$$\operatorname{up}^{(r)}(x) = 2^{T_r}\, F_{\mathrm{gl}}\big(2^r(x+1)\big) \qquad (-1 \leq x \leq 1), \tag{121}$$

from $\operatorname{up}(x) = F_{\mathrm{gl}}(x + 1)$ on $[-1, 1]$; at a dyadic x of depth n it vanishes for $r \geq n + 1$, and outside the support all derivatives vanish. The right-hand sides are evaluated by Equation (119) and any finite formula for F_{bd} at denominator depth at most $n-r$. Thus every entry of every dyadic Taylor jet is explicitly rational, and substituting into Equation (88) or Equation (116) makes every coefficient of the scale polynomials explicit — for example $\mathrm{up}''(\frac{1}{4}) = 2^{T_2} F_{\mathrm{gl}}(5) = 8\,\varepsilon_2 = -8$ by Equation (23). Theorem 7.1 gives the same derivatives as one Thue–Morse sum without the fold.

11.5 Iterated primitives

For an integer $r \geq 1$, repeated integration has an equally simple exact form:

$$\frac{1}{(r-1)!}\int_0^x (x-t)^{r-1} F_{\mathrm{gl}}(t)\,\mathrm{d}t = 2^{\binom{r}{2}}\, F_{\mathrm{gl}}\Big(\frac{x}{2^r}\Big) \qquad (x \geq 0), \tag{122}$$

$$\frac{1}{(r-1)!}\int_0^x (x-t)^{r-1} F_{\mathrm{bd}}(t)\,\mathrm{d}t = 2^{\binom{r}{2}}\, F_{\mathrm{bd}}\Big(\frac{x}{2^r}\Big) \qquad (0 \leq x \leq 1), \tag{123}$$

$$\frac{1}{(r-1)!}\int_{-1}^x (x-t)^{r-1}\, \mathrm{up}(t)\,\mathrm{d}t = 2^{\binom{r}{2}}\, F_{\mathrm{bd}}\Big(\frac{x+1}{2^r}\Big) \qquad (-1 \leq x \leq 1). \tag{124}$$

Proof. The left side of Equation (122) is the r-fold iterated primitive of F_{gl} with vanishing initial data at 0 (Cauchy's formula for repeated integration). Differentiating the right side r times with Equation (22) gives $2^{\binom{r}{2}}\, 2^{T_r}\, 2^{-r^2} F_{\mathrm{gl}}(x) = F_{\mathrm{gl}}(x)$, since $\binom{r}{2} + T_r - r^2 = 0$; its first r derivatives at 0 vanish because F_{gl} is flat there. Two functions with the same rth derivative and the same first r initial data coincide. The second line restricts the first to $[0,1]$, where $F_{\mathrm{gl}} = F_{\mathrm{bd}}$ and $x/2^r \in [0,1]$. The third uses $\mathrm{up}(t) = F_{\mathrm{gl}}(t+1)$ and the substitution $t \mapsto t-1$, which moves the lower endpoint to 0 and the argument to $(x+1)/2^r \in [0,1]$. □

These integrals at dyadic endpoints are therefore rational, with the same recurrence-free evaluations at a larger denominator depth.

12 The reciprocal-integer-base extension

The squared-scale mechanism is not confined to base two. The extension concerns a *different family of distributions*; it is not a statement about the Fabius function at non-dyadic rationals, and it does not assert that the base-b laws share all local analytic properties of F_{bd}.

12.1 The normalized base-b law and its finite arithmetic

For an integer $b \geq 2$ let

$$X_b := (b-1)\sum_{j=1}^{\infty} b^{-j} U_j, \qquad F_{\mathrm{bd}}^{(b)}(x) := \mathbb{P}(X_b \leq x), \tag{125}$$

normalized so that the support is $[0,1]$; $X_2 = X$. The same absolutely convergent series and characteristic-function arguments as in Theorem 2.1 give reflection symmetry $X_b \overset{\mathrm{law}}{=} 1 - X_b$ and a smooth density, flat at both endpoints. Write $Z_b = 2X_b - 1$, with centred moment generating function $M_b(t) = \prod_{j\geq 1} H((b-1)t/b^j)$, even, and with

$$\log M_b(t) = \sum_{r\geq 1} \lambda_{r,b}\, t^{2r}, \qquad \lambda_{r,b} = \frac{(b-1)^{2r}\, 2^{2r-1} B_{2r}}{r\,(2r)!\,(b^{2r}-1)}, \tag{126}$$

by the geometric summation of $\log H$ over the scales b^{-j} exactly as in Theorem 4.2. The raw coefficients have the positive finite formula

$$\boxed{A_m^{(b)} := \frac{\mathbb{E}X_b^m}{m!} = (b-1)^m \sum_{\mathbf{r}\in\mathcal{C}_m} \prod_{i=1}^{\ell(\mathbf{r})} \frac{1}{(r_i+1)!\,(b^{R_i}-1)}\,,} \tag{127}$$

by the gap summation Equation (49) with b in place of 2 and the common scale $(b-1)^m$. Other integer bases do not fill one consecutive integer block with their subset sums, so a Boolean cube sum replaces the single Thue–Morse prefix. For $n \geq 1$, $0 \leq m \leq b^n$ and $\delta \in \{0,1\}^n$ put

$$\Delta_\delta := m - (b-1)\sum_{j=0}^{n-1} \delta_j b^j. \tag{128}$$

Theorem 12.1 (Finite base-b arithmetic)**.** *For these parameters,*

$$\boxed{F_{\mathrm{bd}}^{(b)}\left(\frac{m}{b^n}\right) = \frac{b^{-\binom{n}{2}}}{(b-1)^n} \sum_{\substack{\delta\in\{0,1\}^n \\ \Delta_\delta \geq 1}} (-1)^{|\delta|} \sum_{l=0}^{n} A_l^{(b)} \frac{(\Delta_\delta - 1)^{n-l}}{(n-l)!}\,.} \tag{129}$$

Consequently $F_{\mathrm{bd}}^{(b)}$ is rational at every base-b rational argument.

Proof. Split X_b into its first n coordinates and the independent tail $b^{-n}X_b'$, and apply Theorem 3.1 to the first coordinates, whose weights $(b-1)b^{-j}$, $1 \leq j \leq n$, have product $(b-1)^n b^{-T_n}$. Conditional on the tail, the truncated power at $x = m/b^n$ is $b^{-n^2}\,(\Delta_\delta - X_b')_+^n$ (indexing the subset by the digits δ_j of the weight $(b-1)b^{-(n-j)}$). Since Δ_δ is an integer and $X_b' \in [0,1]$, this is zero for $\Delta_\delta \leq 0$ and an ordinary polynomial for $\Delta_\delta \geq 1$. Reflection replaces $1 - X_b'$ by X_b', and expanding $(\Delta_\delta - 1 + X_b')^n/n!$ gives the inner sum; the scalar factors combine to $b^{T_n - n^2}(b-1)^{-n} = b^{-\binom{n}{2}}(b-1)^{-n}$. □

12.2 Parity-reduced extraction at the base b^{-2}

For $N \geq 1$ let $X_{b,N} = (b-1)\sum_{j=1}^{N} b^{-j}U_j$ and $G_{b,N}(x) := \mathbb{P}(X_{b,N} + \frac{1}{2}b^{-N} \leq x)$, the centred prefix distribution function. For $N \geq n$ and $A = mb^{N-n}$ the box formula gives the finite rational expression

$$G_{b,N}\left(\frac{m}{b^n}\right) = \frac{1}{2^N (b-1)^N b^{\binom{N}{2}} N!} \sum_{\delta\in\{0,1\}^N} (-1)^{|\delta|} \left(2A - 1 - 2(b-1)\sum_{j=0}^{N-1}\delta_j b^j\right)_+^N, \tag{130}$$

in which the positive part is a comparison of explicitly given integers; at a general rational x the same formula holds with $2b^N x$ in place of $2A$.

Theorem 12.2 (Moment-free integer-base extraction)**.** *For $n \geq 1$, $0 \leq m \leq b^n$, $d = \lfloor n/2 \rfloor$, and $R \geq 0$,*

$$\boxed{F_{\mathrm{bd}}^{(b)}\left(\frac{m}{b^n}\right) = \sum_{j=0}^{d} w_{d,j}\left(b^{-2}\right)\, G_{b,n+R+j}\left(\frac{m}{b^n}\right),} \tag{131}$$

with the weights of Equation (95) at $q = b^{-2}$.

Proof. The self-similarity $X_b \overset{\text{law}}{=} ((b-1)U + X_b')/b$ gives

$$\left(F_{\mathrm{bd}}^{(b)}\right)'(x) = \frac{b}{b-1}\left\{F_{\mathrm{bd}}^{(b)}(bx) - F_{\mathrm{bd}}^{(b)}\big(bx - (b-1)\big)\right\}. \tag{132}$$

Iterating this n times writes the nth derivative at x as a finite linear combination of values $F_{\mathrm{bd}}^{(b)}(b^n x - k)$ with integral k; one further differentiation and evaluation at $x = m/b^n$ involves only positive-order derivatives of $F_{\mathrm{bd}}^{(b)}$ at integers, which vanish because integers are endpoints of, or lie outside, the support of the density. Hence the jet of $F_{\mathrm{bd}}^{(b)}$ at m/b^n terminates at order n.

Now write $\eta = b^{-N}$ and use the head–tail identity $F_{\mathrm{bd}}^{(b)}(t) = \mathbb{E}G_{b,N}(t - \frac{\eta}{2}Z_b')$, with Z_b' symmetric and supported in $[-1,1]$. Every knot of the centred finite distribution function is an odd multiple of $\eta/2$, while m/b^n is an integer multiple of η for $N \geq n$; the averaging interval of half-width $\eta/2$ therefore lies in one polynomial cell, and the fact that not every odd multiple is a knot causes no difficulty. Theorem 7.2 with $W = Z_b'$ and the terminating jet shows that $G_{b,N}(m/b^n)$ is an exact polynomial of degree at most d in $(\eta/2)^2$, hence in b^{-2N}, with constant term $F_{\mathrm{bd}}^{(b)}(m/b^n)$. Geometric constant extraction with $q = b^{-2}$ at the depths $n + R + j$ proves Equation (131), and substituting Equation (130) shows rationality once more. □

For example, the explicit sums give

$$F_{\mathrm{bd}}^{(3)}\left(\tfrac{1}{9}\right) = \frac{7}{576}, \quad F_{\mathrm{bd}}^{(3)}\left(\tfrac{1}{27}\right) = \frac{1}{6912}, \quad F_{\mathrm{bd}}^{(4)}\left(\tfrac{1}{16}\right) = \frac{1}{240}, \quad F_{\mathrm{bd}}^{(4)}\left(\tfrac{1}{64}\right) = \frac{1}{51840}.$$

At $b = 2$ the cube sums collapse to the dyadic prefix and every formula of the preceding sections is recovered. For $b > 2$ some of the highest jets can vanish for additional geometric reasons, so only the degree *upper bound* d is asserted, not the exact-degree statement of Theorem 7.7. The existence of exact finite extraction is caused by two structures together — grid-aligned finite-convolution knots and an even centred tail — and neither alone would give the degree bound.

13 Worked exact values and the retained verifier

13.1 Universal coefficients

The first normalized and raw moments, all obtained independently from the composition, partition, and finite-prefix formulas, are

k	a_k	$c_k = \mathbb{E}Z^{2k}$	e_k	λ_k
0	1	1	1	—
1	1/18	1/9	−1/18	1/18
2	19/16200	19/675	31/16200	−1/2700
3	583/42865200	583/59535	−2347/42865200	1/178605
4	132809/1311675120000	132809/32531625	1904369/1311675120000	−1/9639000

These numbers are universal: the dyadic numerator changes only the outer polynomial evaluation. The raw coefficients begin $A_1 = \frac{1}{2}$, $A_2 = \frac{5}{36}$, $A_3 = \frac{1}{36}$, $A_4 = \frac{143}{32400}$, $A_5 = \frac{19}{32400}$, and the reciprocal powers $F_{\mathrm{bd}}(2^{-n}) = 2^{-\binom{n}{2}} A_n$ for $n = 0, \ldots, 7$ are

$$1, \ \tfrac{1}{2}, \ \tfrac{5}{72}, \ \tfrac{1}{288}, \ \tfrac{143}{2073600}, \ \tfrac{19}{33177600}, \ \tfrac{1153}{561842749440}, \ \tfrac{583}{179789679820800}.$$

13.2 Two and three samples

At $x = \frac{1}{4}$ ($n = 2$, $d = 1$) the explicit spline sums are $G_2(\frac{1}{4}) = \frac{1}{16}$ and $G_3(\frac{1}{4}) = \frac{13}{192}$ — the numerator of the latter is $3^3 - 1^3 = 26$ over $2^{T_3}3! = 384$ — so

$$F_{\mathrm{bd}}\left(\tfrac{1}{4}\right) = \frac{-G_2(\frac{1}{4}) + 4G_3(\frac{1}{4})}{3} = \frac{5}{72}, \qquad G_N\left(\tfrac{1}{4}\right) = \frac{5}{72} - \frac{1}{9}4^{-N} \ \ (N \geq 2). \tag{133}$$

Likewise $G_3(\frac{3}{8}) = \frac{97}{384}$, $G_4(\frac{3}{8}) = \frac{389}{1536}$, and $F_{\mathrm{bd}}(\frac{3}{8}) = (4G_4 - G_3)/3 = \frac{73}{288}$; and $G_3(\frac{1}{8}) = \frac{1}{384}$, $G_4(\frac{1}{8}) = \frac{5}{1536}$, $F_{\mathrm{bd}}(\frac{1}{8}) = \frac{1}{288}$ with $G_N(\frac{1}{8}) = \frac{1}{288} - \frac{1}{18}4^{-N}$. At depth four three levels suffice:

$$G_4\left(\tfrac{1}{16}\right) = \frac{1}{24576}, \qquad G_5\left(\tfrac{1}{16}\right) = \frac{121}{1966080}, \qquad G_6\left(\tfrac{1}{16}\right) = \frac{6331}{94371840},$$

$$F_{\mathrm{bd}}\left(\tfrac{1}{16}\right) = \frac{G_4 - 20G_5 + 64G_6}{45} = \frac{143}{2073600}, \tag{134}$$

and here the polynomial in $t = 4^{-N}$ is genuinely quadratic:

$$G_N\big(\tfrac{1}{16}\big) = \frac{143}{2073600} - \frac{5}{648}\,t + \frac{248}{2025}\,t^2, \qquad G_N\big(\tfrac{3}{16}\big) = \frac{46657}{2073600} - \frac{67}{648}\,t - \frac{248}{2025}\,t^2 \quad (N \geq 4). \tag{135}$$

At $x = \frac{5}{16}$, with $G_4 = \frac{1205}{8192}$, $G_5 = \frac{289799}{1966080}$, $G_6 = \frac{13917509}{94371840}$, the weighted sum $\frac{1}{45}G_4 - \frac{4}{9}G_5 + \frac{64}{45}G_6$ is exactly $\frac{305857}{2073600}$. No previously known value of F_{bd} enters any of these extractions.

13.3 A table of exact values

All entries of Table 2 were matched by the master, partition, both binary, determinant, finite-cube and quarter-base formulas in exact rational arithmetic. Reflection supplies the remaining values on each grid: $F_{\mathrm{bd}}((16-m)/16) = 1 - F_{\mathrm{bd}}(m/16)$, and the bump values follow from $\mathrm{up}(m/16) = F_{\mathrm{bd}}((16-|m|)/16)$, for example $\mathrm{up}(\frac{11}{16}) = F_{\mathrm{bd}}(\frac{5}{16})$.

Table 2: Selected exact values, fractions reduced.

x	$F_{\mathrm{bd}}(x)$	x	$F_{\mathrm{bd}}(x)$
1/2	1/2	1/4	5/72
1/8	1/288	3/8	73/288
1/16	143/2073600	3/16	46657/2073600
5/16	305857/2073600	7/16	777743/2073600
1/32	19/33177600	13/32	10393219/33177600
21/64	482385079229/2809213747200	37/128	4121049919811/35957935964160

Two further values: $F_{\mathrm{bd}}(1/1024) = 134926369/1246394851358539387238350848000 \approx 1.082533 \cdot 10^{-22}$, where the positive formula avoids subtracting large alternating terms, and $F_{\mathrm{bd}}(173/4096)$ of Equation (67). The decimal displays are for orientation; the fractions are the actual outputs.

13.4 Operation counts and exact arithmetic

For the consecutive-depth quarter formula with $R = 0$ the raw integer sums contain $m(2^{d+1} - 1)$ integer-power summands before any use of symmetry or cancellation identities, and the largest depth is $n + \lfloor n/2 \rfloor$; reducing the fraction first, and replacing x by $1 - x$ when that is closer to zero, decreases the count. The block formula has $\mathrm{wt}_2(m)$ outer terms and needs coefficient indices at most $\lfloor n/2 \rfloor$ regardless of the numerator; the telescope has the same number of outer terms with raw coefficients of index up to n. The signs in $\mathcal{S}_N(A)$ can produce severe cancellation between very large integers: compute the numerator exactly, divide only after summation, and combine the interpolation weights as exact fractions. The bounded norm of the weight row (Theorem 8.5) does not make a low-precision evaluation of the individual sums safe. None of this asserts that a nonrecursive formula must outperform a well-designed recurrence.

13.5 What the retained verifier checks

The retained program `verify_recurrence_free_dyadic_values.py` uses only the Python standard library and `fractions.Fraction`; every comparison is an exact equality, with no floating-point tolerance. It implements, as separate finite evaluators, the master formula with composition and with partition coefficients, the raw-moment master formula, the block formula (bounded and signed), the telescope, the quarter-base row at consecutive and at arbitrary depths, the finite-cube formula, the bordered determinant (by exact Bareiss elimination), the certified rounding, the half-base row with arbitrary shift α and its closed form, the direct density extraction, and the integer-base formulas; the only recursive object, the triangular moment recurrence, is isolated and used solely as a cross-check. Its recorded run (`verification_results.json`) performs the checks of Table 3.

Table 3: Exact checks performed by the retained verifier at maximal depth 8.

Check	Cases
Coefficients a_k by compositions, partitions and the isolated oracle, $0 \leq k \leq 12$; recovery from finite prefixes at two starting depths, multinomial against power-sum prefix coefficients, both reciprocal expansions, $(-1)^k e_k > 0$ and $\sum_j a_j e_{k-j} = [k = 0]$, $0 \leq k \leq 8$; λ_r by the Bernoulli and the Bernoulli-free formula, $r \leq 8$; raw A_m by compositions, partitions and the dictionary Equation (30), $m \leq 8$; every printed constant	39
Weight rows $d \leq 8$: product, q-Pochhammer, Gaussian-binomial and integer forms agree; annihilation identities; absolute row sum and generating polynomial at three bases; the bound < 2	9 rows
Every representation $m/2^n$, $0 \leq n \leq 8$: master (three coefficient routes), block, signed block, telescope, quarter-base, determinant (and finite-cube for $n \leq 6$) agree; reflection; refinement $(m, n) \mapsto (2m, n+1)$; integrality against D_n and D'_n	1,033
Shifted-depth and arbitrary-depth extraction at selected points	84
Signed extension $F_{\mathrm{gl}}(a/2^n)$ by blocks against the fold, $0 \leq a \leq 3 \cdot 2^n$, $n \leq 6$; integer and half-integer values	388
Exact shrinking-cell identity at five offsets, grids through depth 5, three depths; scale polynomials through depth 6 at three depths; exact degree at every reduced interior point; the four printed polynomials	1,005
Dyadic derivatives by the differentiated master formula against the dilation law, orders $0 \leq r \leq n+1$, $n \leq 5$	434
Certified rounding, depths 1–3	17
Half-base row with $\alpha \in \left\{0, \frac{1}{2}, 1, -2, \frac{2}{3}\right\}$, and its closed form	65
Rvachev values by direct density extraction (two starting depths), the finite fold identity, the density law Equation (116), $\mathrm{up}''(\frac{1}{4}) = -8$	126
Odd-depth and reciprocal-power formulas; the eight anchor values	22
Integer bases $b = 2, 3, 4, 5$: extraction against the finite arithmetic formula and at a shifted depth, reflection, cube-sum evaluations	235

The grid cases are numerator–depth pairs, not distinct rational points: unreduced representations are intentionally included. To reproduce the recorded run:

```
python verify_recurrence_free_dyadic_values.py --max-depth 8 --json
    verification_results.json
```

The exhaustive sums grow exponentially with the requested depth. The tests check the implementations and the indexing; they are not a substitute for the proofs, nor a claim that any of this has been formalized in Lean.

14 How the alternatives compare, and what remains

14.1 Closed form versus evaluation algorithm

An explicit finite formula may be evaluated by loops, memoization, or an exact linear-algebra algorithm without ceasing to be a closed form; what is excluded is a definition that leaves an auxiliary constant specified only through earlier unknown constants. Storing previously evaluated a_k is optional caching, whereas defining a_k only by Equation (51) would fail the criterion.

The families are algebraically equivalent descriptions but not identical algorithms. The composition sum has 2^{k-1} positive terms, the partition sum one signed term per integer partition, the finite-prefix representation only weak compositions and rational Gaussian weights. Once coefficients are available, the block formula has at most $(d+1)(n-d+1)$ polynomial terms. The quarter-base double sum is the most direct answer with no named coefficient sequence, at the price of $m(2^{d+1}-1)$ signed powers and enormous intermediate integers. The determinant supplies a common denominator immediately, but its permutation expansion is not the preferred numerical algorithm. Large powers and cancellation, bit lengths of rational intermediates, coefficient reuse, and whether one value or a whole grid is wanted all matter more than the number of summands.

14.2 Further deductions that need no new conjecture

A finite subset of the cutoff coefficients can vanish at special points; whenever the nonzero orders are known one may interpolate only those monomials rather than the complete degree space. The exact-degree result guarantees that the top order does not vanish at reduced points but does not exclude gaps at lower orders, which suggests point-adapted extraction rows whose validity is proved by the corresponding finite moment equations. The local inverse lemma applies more generally whenever a smooth law is an independent sum of a finite spline head and a scaled tail, the tail's support fits in one polynomial cell at the evaluation point, and the relevant jet of the limit terminates: symmetry removes the odd cutoff powers, other coefficient zeros can remove further powers, and the search for additional exact formulas separates into a geometric question about knot cells and an algebraic one about the nonzero jet orders. The integer-base theorem is one realization of that separation. Open beyond this document: a formalization of the general polynomial law (Section D), and the sharp denominator arithmetic of [5] against the elementary bounds D_n, D'_n.

14.3 Conclusion

Every dyadic value in question admits several closed forms in the strict finite sense. The hidden moment recurrences of the classical formula can be eliminated by explicit multiplicity partitions, by positive ordered compositions, by finite-prefix interpolation, or by an integer determinant. Binary digits provide two finite compressions with no long signed prefix. Most importantly, centred finite convolutions have an exact polynomial dependence on the squared tail scale at dyadic arguments; removing that polynomial by geometric interpolation gives Equation (6), a moment-free $q=\frac{1}{4}$ formula requiring only $\lfloor n/2 \rfloor + 1$ spline levels, and the shrinking-cell theorem explains why the quarter base replaces the half base and why that number of samples is the least possible for a grid-wide rule. Reflection and folding cover every Rvachev and signed Fabius value, and the same finite algebra extends to all derivatives, iterated primitives, and reciprocal-integer-base laws. Infinite products and probability models appear in the proofs; neither infinite limits nor auxiliary recurrences survive in the evaluation formulas.

A Compact exact evaluators

The following self-contained implementations use only integer powers, factorials, binary digit counts, and exact fractions. The first is the principal formula Equation (6); the second is the block formula Equation (61) with the composition coefficients Equation (47), valid for the signed extension at every nonnegative numerator. Python 3.10 or later suffices; pass exact rational inputs such as `Fraction(173, 4096)` rather than decimal floats, whose exact binary value may have a much larger denominator than intended.

```
from fractions import Fraction
from functools import lru_cache
from math import factorial, prod


def fabius_quarter(m: int, n: int) -> Fraction:
    """Exact bounded F(m / 2**n) for 0 <= m <= 2**n by the quarter-base row."""
    if n < 0 or not 0 <= m <= 2**n:
        raise ValueError("require n >= 0 and 0 <= m <= 2**n")
    while n > 0 and m % 2 == 0:      # reduce: fewer levels
        m //= 2; n -= 1
    if n == 0:
        return Fraction(m)
    d = n // 2
    total = Fraction(0)
    for j in range(d + 1):
        weight = Fraction((-1)**(d - j) * 4**(j*(j+1)//2),
                          prod(4**r - 1 for r in range(1, j+1))
                          * prod(4**r - 1 for r in range(1, d-j+1)))
        N, A = n + j, m * 2**j
        power_sum = sum((-1)**h.bit_count() * (2*A - 2*h - 1)**N
                        for h in range(A))
        total += weight * Fraction(power_sum, 2**(N*(N+1)//2) * factorial(N))
    return total


def compositions(k: int):
    if k == 0:
        yield (); return
    for mask in range(1 << (k - 1)):
        cuts = [0] + [j for j in range(1, k) if (mask >> (j-1)) & 1] + [k]
        yield tuple(v - u for u, v in zip(cuts, cuts[1:]))


@lru_cache(None)
def coefficient(k: int) -> Fraction:
    """a_k = [t^(2k)] M(t) by the positive composition formula."""
    total = Fraction(0)
    for parts in compositions(k):
        term, suffix = Fraction(1), k
        for r in parts:
            term /= factorial(2*r + 1) * (4**suffix - 1); suffix -= r
        total += term
    return total


def fabius_global_blocks(a: int, n: int) -> Fraction:
    """Signed global F_gl(a / 2**n), every a >= 0, by the binary block formula.""
```

```
    "
    if a < 0 or n < 0:
        raise ValueError("a and n must be nonnegative")
    total = Fraction(0)
    for i in range(n + 1):
        if not (a >> i) & 1:
            continue
        sign = -1 if (a >> (i + 1)).bit_count() % 2 else 1
        K = (1 << i) + 2 * (a % (1 << i))
        inner = sum((coefficient(k) * 4**(i*k)
                     * Fraction(K**(n-i-2*k), factorial(n-i-2*k))
                     for k in range((n-i)//2 + 1)), Fraction(0))
        total += sign * 2**(i*(i+1)//2) * inner
    return total / 2**(n*(n+1)//2)


def rvachev_up(a: int, n: int) -> Fraction:
    """Exact up(a / 2**n) on the whole dyadic grid."""
    return Fraction(0) if abs(a) > 2**n else fabius_quarter(2**n - abs(a), n)


assert fabius_quarter(1, 4) == Fraction(143, 2073600)
assert fabius_global_blocks(173, 12) == fabius_quarter(173, 12)
assert fabius_global_blocks(3, 0) == -1 and rvachev_up(3, 2) == Fraction(5, 72)
```

B A direct finite definition of every index set

To remove any ambiguity about the word “finite”, all auxiliary indexing reduces to bounded integer tuples. A composition of $k \geq 1$ is a length $1 \leq \ell \leq k$ and a tuple $(r_1, \ldots, r_\ell) \in \{1, \ldots, k\}^\ell$ with $r_1 + \cdots + r_\ell = k$; the empty tuple is the only composition of 0. A multiplicity vector of weight k lies in $\prod_{r=1}^{k} \{0, 1, \ldots, \lfloor k/r \rfloor\}$ and satisfies $\sum_r r v_r = k$; a raw profile of weight m adds $v_0 \in \{0, \ldots, m\}$ with $v_0 + 2\sum_r r v_r = m$. Binomial coefficients are factorial quotients in their stated ranges; Bernoulli numbers are the finite sums Equation (39); every Thue–Morse sign is the digit product Equation (3); every determinant is the permutation sum Equation (109); every q-Pochhammer symbol in a value formula is the finite product Equation (11); the digit data Equation (60) and Equation (69) are floor and remainder operations. There is no unstated recursive sequence anywhere in the evaluation layer.

C Notation

Every symbol is normative from the corpus notation catalogue where an entry exists (`rvachev-up`, `fabius-bounded`, `fabius-global`, `thue-morse`, `q-pochhammer`, `gaussian-binomial`); the remaining symbols are document-local and listed here.

Symbol	Meaning
F_{bd}, up, F_{gl}	bounded distribution function of X; density of $Z = 2X - 1$; signed global extension Equation (20)
X, Z; U_j, V_j	the random sums Equations (12) and (13); uniforms on $[0,1]$ and $[-1,1]$
X_N, Z_N, Y_N	uncentred and centred prefixes; $Y_N = X_N + 2^{-N-1}$
H_N, G_N, $G_{N,\alpha}$, u_N	distribution functions of X_N, Y_N, $X_N + \alpha 2^{-N}$; density of Z_N
H, M, M_N, M_X	$\sinh t/t$; moment generating functions of Z, Z_N, X
a_k, c_k, A_m	$\left[t^{2k}\right] M$, $\mathbb{E}Z^{2k}$, $\mathbb{E}X^m/m!$
λ_r, β_r^*, e_k	coefficients of $\log M$, of $\log M_X - z/2$, of $1/M$
$\Phi_r(v)$, $C_{n,N}(A)$, $a_{k,N}$	$[t^r]\, \mathrm{e}^{vt} M$; $\mathbb{E}(A + Z_N)^n/n!$; $\left[t^{2k}\right] M_N$
T_r, d, ε_h	$r(r+1)/2$; $\lfloor n/2 \rfloor$; Thue–Morse sign $(-1)^{\mathrm{wt}_2(h)}$
$\mathcal{S}_N(A)$, t_j, L_d, $B_{n,m}$, D_n, D'_n	integer power sum Equation (4); determinant data Section 10
$\mathcal{C}_k$, R_i, $\mathcal{P}_k$	compositions, suffix sums, multiplicity profiles Section 1.5
$w_{d,j}(q)$, $w_{d,j}$	geometric extraction weights Equation (95); at $q = \frac{1}{4}$
P_x, Q_x	dyadic Taylor polynomial; scale polynomial Equation (88)
$J(m)$, p_j, r_j, K_j, η_j; b_i, s_i	block data Equation (60); telescope data Equation (69)
X_b, $F_{\mathrm{bd}}^{(b)}$, $G_{b,N}$, $A_m^{(b)}$, $\lambda_{r,b}$	the base-b objects of Section 12
a_r^{vol}	the canonical volume's reciprocal-product coefficients, $= (-1)^r e_r$

D Formalization register

The results of this document are human proofs. The repository's Lean development [1] contains counterparts for several of the ingredients, which the canonical volume's register [2] already lists in the density normalization; the rows below state the correspondence for the objects as they appear here. Every declaration name was checked to resolve in the repository on the day of consolidation. Statuses: CROSSWALKED, a declaration states the same fact (with the translation stated); INSTANCE, kernel-verified for one point or one mode; ABSENT, no declaration found.

Table 4: Formalization register.

Result here	Lean declaration(s)	Status
Equation (81): the centred finite spline as a Thue–Morse truncated-power sum	`Fabius.fabiusUniformSpline` (`FabiusUniformSpline`), defined as that sum in the centred-CDF variable; `Fabius.ProbabilityRepresentation.fabiusUniformSpline_eq_centeredPartialCDF`	CROSSWALKED
Equation (13), Theorem 2.1: up is the density of $\sum_k 2^{-k} V_k$	`Fabius.ProbabilityRepresentation.rvachevUp_eq_weightedSum_probability`, `...geometricUniformDensity_one_half_eq_rvachevUp`	CROSSWALKED
Equation (53), Equation (44): the reciprocal and moment coefficients in Bell form	`Fabius.rvachevReciprocalMomentRat`, `..._eq_completeBellPolynomial`, `Fabius.binomialConv_rvachevRawMomentRat_reciprocal` (`RvachevMomentAppell`): $a_r^{\mathrm{vol}} = (-1)^r e_r$ is Lean-computable	CROSSWALKED; positivity Theorem 7.6: ABSENT
Theorem 7.2 at scale 1	`Fabius.rvachevDeconvolvedPolynomial`, `Fabius.rvachevAppellPolynomialRat`, `Fabius.integral_eval_rvachevAppellPolynomial_sub_mul_rvachev` (`RvachevMomentAppell`)	CROSSWALKED at scale 1; the dilation and the cell identity: ABSENT

Result here	Lean declaration(s)	Status
Theorem 2.2: dilation law and the terminating dyadic jet	Fabius.deriv_rvachevUp, Fabius.iteratedDeriv_rvachevUp_eq_thueMorse_sum (Differential); Fabius.iteratedDeriv_rvachevUp_dyadic_eq_zero, Fabius.dyadic_depth_eq_max_nonzero_iteratedDeriv (DyadicDerivativeFiltration); Fabius.iteratedDeriv_fabiusReal_dyadicGrid_eq_ite, Fabius.abs_iteratedDeriv_fabiusReal_dyadicGrid_of_odd (BoundedDerivatives)	CROSSWALKED (the top derivative 2^{T_n} at odd numerators is Theorem 7.7's nonvanishing)
Equation (89): the cell identity at $x = \frac{1}{4}$	Fabius.fabiusUniformSpline_quarter_eq_quadratic, Fabius.reportFiniteFabiusApproximant_quarter_eq_quadratic (QuarterSplineLocalPolynomial): $\frac{5}{72} + z + 4z^2 - \frac{4}{9}\,4^{-n}$ in the volume's level index $n = N - 1$	INSTANCE; Theorems 7.3 and 7.4 at general depth: ABSENT
Theorem 8.2: the weights, their closed forms, row polynomial and moment identities	Fabius.geometricLagrangeWeight, . . . _eq_product (GeometricLagrangeWeights); Fabius.quarterLagrangeWeight_eq_qPochhammer, . . . _eq_qBinomial (GeometricLagrangeQBinomial); Fabius.sum_quarterLagrangeWeight_eq_one, . . . _mul_pow_eq_zero, Fabius.quarterLagrangeWeightPolynomial_eq_forwardRichardson	CROSSWALKED
Theorem 1.1, Theorem 8.3: the row applied to the spline values	Fabius.quarterLagrange_rvachevFourierProduct_eq_gaussian_tsum (RvachevQBinomialFilter) applies the same row to the infinite product on the Fourier side	neighbouring only; the theorem: ABSENT, pending the law
Theorem 3.2, Theorem 4.6, Theorem 4.4, Theorem 5.1, Theorem 5.4, Theorem 6.1, Theorem 6.2, Theorem 10.1, Theorem 10.5, Theorem 9.1, Theorem 11.1, Section 11.5, Section 12	—	ABSENT

A formalization of the general polynomial law would follow the five-step plan recorded in the canonical volume (finite density and knot geometry, head–tail factorization at finite level, dilated deconvolution and the cell identity, assembly with the jet termination); the master identity and the coefficient formulas of Sections 3 and 4 need only finite algebra over the existing `RvachevMomentAppell` coefficients and the box formula Theorem 3.1.

E Provenance of the consolidation

E.1 What was merged

Three independently written articles arrived on 2026-09-04 and were filed the same day, as separate members beside the canonical volume, in the subgroup `inverse-and-sampling/dyadic-up-extraction/`. All three answer the same question — exact evaluation of $F_{\mathrm{bd}}(m/2^n)$ and $\mathrm{up}(m/2^n)$ with no auxiliary quantity defined by a moment recurrence or a limiting process — and all three reach the quarter-base formula through the same mechanism. They were merged editorially into this document on 2026-09-04: one statement of each result, every proof completed, the four letters for the numerator, the three names for the moment coefficients, the three names for the reciprocal coefficients, and the two spline indexings reconciled once, and every printed rational checked by the retained verifier. The three directories and their retained arrival PDFs were deleted after a residue audit; git history is the archive.

E.2 The absorbed sources and what each contributed

S1: ***Recurrence-Free Closed Forms for Dyadic Fabius and Rvachev Values***
(`fabius_dyadic_closed_forms/`, 1,441 lines, 22 pp.). The Taylor-integral proof of the master identity for the signed extension (Theorem 3.3); the block formula in integer form and its validity at every numerator (Theorem 5.1); the Bernoulli-free cumulant formula (Theorem 4.3); the finite-cube value formula and the arbitrary-prefix weights (Theorem 6.2); the $(d+2)$-bordered determinant (Theorem 10.2); the Prouhet route to the degree drop (Theorem 7.5); the density splines $S_N = G_{N-1}$; the examples $F_{\mathrm{bd}}(1/1024)$, $F_{\mathrm{bd}}(173/4096)$ and the three-sample extraction at $\frac{5}{16}$.

S2: ***Recurrence-Free Dyadic Values of the Fabius and Rvachev Functions***
(`fabius_dyadic_closed_forms-2/`, 1,456 lines, 26 pp.). The organization of Sections 1 to 3; the exact shrinking-cell theorem (Theorem 7.3); the exact degree and the minimal-sample theorem (Theorems 7.7 and 8.4); the bound < 2 on the row (Theorem 8.5); the finite-prefix coefficients and their extraction (Section 6); the differentiated master formula (Theorem 7.1); the composition denominator bound, the prefix error and the certified rounding (Theorems 10.3 and 10.5); the integer-base cumulants and the centred base-b prefix; the example $21/64$ and the quadratic scale polynomials at $\frac{1}{16}$, $\frac{3}{16}$.

S3: ***Recurrence-Free Dyadic Formulae for the Fabius and Rvachev Functions***
(`fabius_rvachev_recurrence_free_closed_forms/`, 1,681 lines, 23 pp.). The raw-moment master identity and the raw composition and partition formulas (Equation (33) and Theorems 4.4 and 4.6); the odd-depth formula Equation (58); the binary telescope (Theorem 5.4); the $(d+1)$-bordered determinant and D_n (Theorem 10.1); the general finite-jet deconvolution lemma (Theorem 7.2); the row's generating polynomial and the numerical limits of the row norm; the half-base identity with arbitrary shift (Theorem 9.1); the direct density extraction (Theorem 11.1); the iterated primitives (Section 11.5); the finite base-b arithmetic and extraction (Theorems 12.1 and 12.2); the index-set appendix.

E.3 Conventions reconciled

S1 wrote the numerator as a, the normalized moments as b_k, the reciprocal coefficients as α_r, the signed extension as Φ, and worked with density splines $S_N(x) = p_N(x-1)$, so that its levels $n+1, \ldots, n+1+d$ are the levels $n, \ldots, n+d$ of G_N here; S2 wrote m, a_k, e_k, Θ, G_N, which this document adopts; S3 wrote a, C_j, R_j, $\mathcal{F}$, $\mathsf{H}_N = G_N$ and the cumulant coefficients λ_r (S1's β_r), adopted here, with S3's raw β_r renamed β_r^*. S1's finite Bernoulli sum and S2/S3's are the same formula. The three common-denominator statements were two: D_n (S1, S3) and D_n' (S2), both kept. Where the sources stated a result without proof — the Bernoulli-free cumulant formula, the coefficient-only prefix formula, the generating polynomial of the row, the primitives — a proof was written; where a proof was sketched — the integer-base extraction in S2, the raw-to-centred dictionary, the determinant's Schur step — it was completed. No mathematical claim of any source was found to be false; the only corrections were indexing conventions, which are recorded above.

E.4 Status of the claims

Every theorem, proposition, lemma and corollary carries a proof. Every displayed rational number, weight row, polynomial coefficient and example value is asserted by the retained verifier (Section 13.5), whose recorded run passed with the counts of Table 3. The verifier merges the three retired programs of the sources (`verify_formulas.py`, `verify_closed_forms.py`, `verify.py`) and adds the checks introduced by the consolidation: the raw-to-centred dictionary, the Bernoulli-free cumulants, the second reciprocal expansion, the arbitrary-depth weights, the closed half-base form, the density law, the derivative formula against the dilation law, and every constant printed here. No Lean formalization of the new material is claimed; Section D is the boundary.

References

[1] V. Reshetnikov and contributors, *The Fabius Function and Rvachev's Up-Function: Exact arithmetic, probability, Fourier analysis, and corrected sharp asymptotics*, primary exposition of the `ProveIt` repository, `Analysis/FabiusFunction/docs/Fabius_Function_and_Rvachev_Up/`, together with the Lean development under `Analysis/FabiusFunction/`. `https://github.com/VladimirReshetnikov/ProveIt`.

[2] V. Reshetnikov, *Exact Dyadic Extraction of Rvachev's Up-Function from Finite Sinc-Product Splines*, canonical consolidated volume of the subgroup `drafts/inverse-and-sampling/dyadic-up-extraction/Dyadic_Up_Extraction/`, 3 September 2026.

[3] V. Reshetnikov, *Geometric q-Fabius Frontiers*, consolidated volume `drafts/exponents-and-q-series/geometric_q_fabius_frontiers/`, Part III (the finite sinc-product splines and the quarter-base Richardson combination).

[4] J. Arias de Reyna, *An infinitely differentiable function with compact support: Definition and properties*, English translation (2017) of the paper in *Rev. Real Acad. Ciencias Exactas Fís. Natur. Madrid* **76** (1982), 21–38. `https://arxiv.org/abs/1702.05442`.

[5] J. Arias de Reyna, *Arithmetic of the Fabius function*, *Integers* **18** (2018), Paper A51, 20 pp. `https://math.colgate.edu/~integers/s51/s51.pdf`; preprint `https://arxiv.org/abs/1702.06487`.

[6] J. K. Haugland, *Evaluating the Fabius function*, arXiv:1609.07999, version 2 (2020; first version 2016). `https://arxiv.org/abs/1609.07999`.

[7] V. Reshetnikov, *Conjectured formula for the Fabius function*, Mathematics Stack Exchange, question posted 5 July 2019. `https://math.stackexchange.com/questions/3283519/conjectured-formula-for-the-fabius-function`.

[8] I. Pinelis, answer to *A non-recursive, explicit formula for the Fabius function*, MathOverflow, January 2020. `https://mathoverflow.net/questions/351004/a-non-recursive-explicit-formula-for-the-fabius-function`.

[9] NIST Digital Library of Mathematical Functions, §4.22 (infinite products and partial fractions, Eq. 4.22.1), §17.2 (q-calculus), §24.2 (Bernoulli numbers). `https://dlmf.nist.gov/`.